Article

Modular CNN–Transformer Specialists for Multi-Source 12-Lead ECG Rhythm and Pathology Classification

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Abstract

Multi-source electrocardiogram (ECG) classification involves heterogeneous labels and evaluation populations. We designed and evaluated modular convolutional neural network (CNN)–Transformer specialists for 10-second 12-lead ECGs, using separately fitted rhythm and pathology backbones with 58.35 million parameters each. Primary6 (P6) reports six rhythm and conduction statements, whereas Primary4 (P4) comprises four binary pathology panels. In selected retrospective evaluations, P6 achieved 95.19% top-1 accuracy and 80.09% mean binary F1 on 1,643 ECGs. P4 achieved 80.23% mean binary accuracy and 79.35% macro-F1 across 962 ECG–label observations from 733 distinct ECGs. Additional analyses examined cohort selection, morphology contributions, and the trade-off between classification errors and withheld decisions. In a separate patient-held-out evaluation within CODE following supervised domain adaptation, a five-endpoint model achieved 60.27% macro-F1 on 4,921 ECGs from 3,891 patients, but did not meet all pre-specified criteria for replacing the released models. These findings describe performance within the evaluated populations rather than a complete clinical diagnostic workflow. Incomplete historical training records limit confirmation of independence across all partitions, and independent institutional validation remains pending.

Keywords: 12-lead electrocardiogram; deep learning; CNN–Transformer; multi-specialist architecture; arrhythmia classification; myocardial infarction; probability calibration; predictive uncertainty; explainable artificial intelligence

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1. Introduction

Artificial intelligence (AI) methods for automated electrocardiogram (ECG) interpretation, including convolutional neural networks (CNNs), have been evaluated on single-lead ambulatory recordings and large 12-lead datasets [1,2]. Multi-label studies and benchmarks support the assessment of rhythm, conduction, and other ECG statements, although the available endpoints and annotation procedures differ across sources [3–6]. These differences affect the interpretation of performance estimates, particularly when

class support varies or uncertain predictions are deferred [7]. Figure 1 summarizes the study workflow, from heterogeneous ECG inputs and signal harmonization to separate rhythm and pathology reporting profiles.

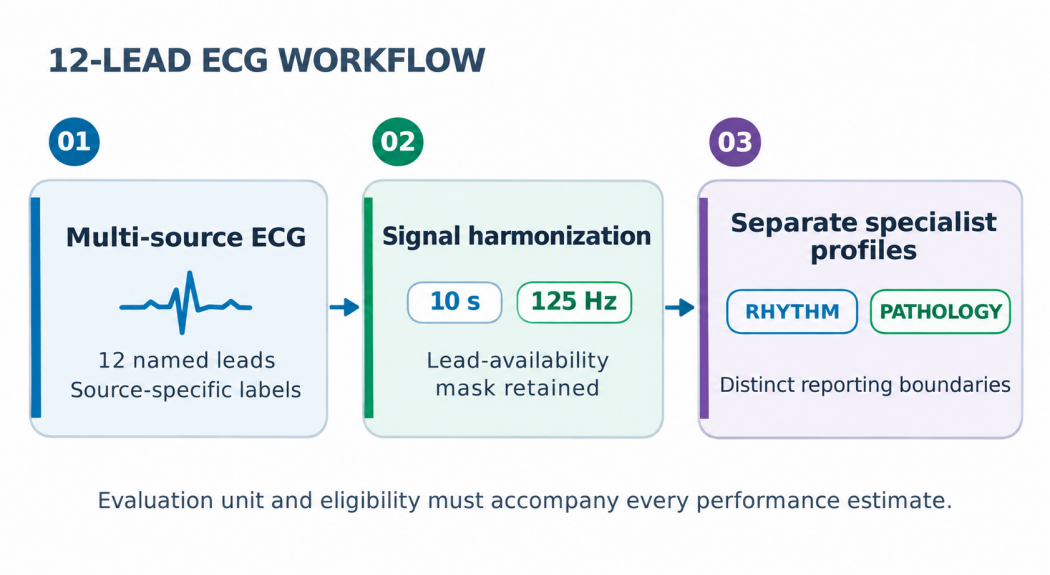


Figure 1. Multi-source 12-lead ECG workflow mapping. The diagram displays the sequential transition from heterogeneous raw inputs to the separated rhythm and pathology specialist boundaries, illustrating the required physical harmonization constraints.

The present study evaluates statements assigned to an entire ECG. A label attached to an individual heartbeat identifies a local event, whereas a longitudinal endpoint concerns a patient over time. Sensitivity and positive predictive value for individual beats describe a different task from classification of a ten-second multi-label ECG. Figure 2 distinguishes these reference levels and places the study at the ECG level. It illustrates the scope of the evaluation without comparing diagnostic performance.

Datasets differ in both their labeling procedures and the findings they record [6]. PTB-XL provides structured SCP-ECG statements, including likelihood information for diagnostic statements [5]. A common output index can therefore combine labels with different reference information. In particular, a finding that was not annotated cannot be assumed equivalent to one explicitly labeled absent. Interpreting performance requires the mapped endpoint, the known reference states, and the observation unit to be considered together.

High aggregate accuracy can conceal poor recall for an uncommon finding. Classwise confusion counts, unweighted binary F1 summaries, and precision–recall curves show aspects of performance that an overall accuracy value may obscure. Each summary must be interpreted for the evaluated population and endpoint prevalence. Focal loss reduces the contribution of well-classified training examples [8]; it does not, by itself, ensure unbiased evaluation, calibrated probabilities, or predictive values that generalize to another population. These properties require assessment in an explicitly defined population. Restricting the evaluation denominator alone does not establish clinical reliability.

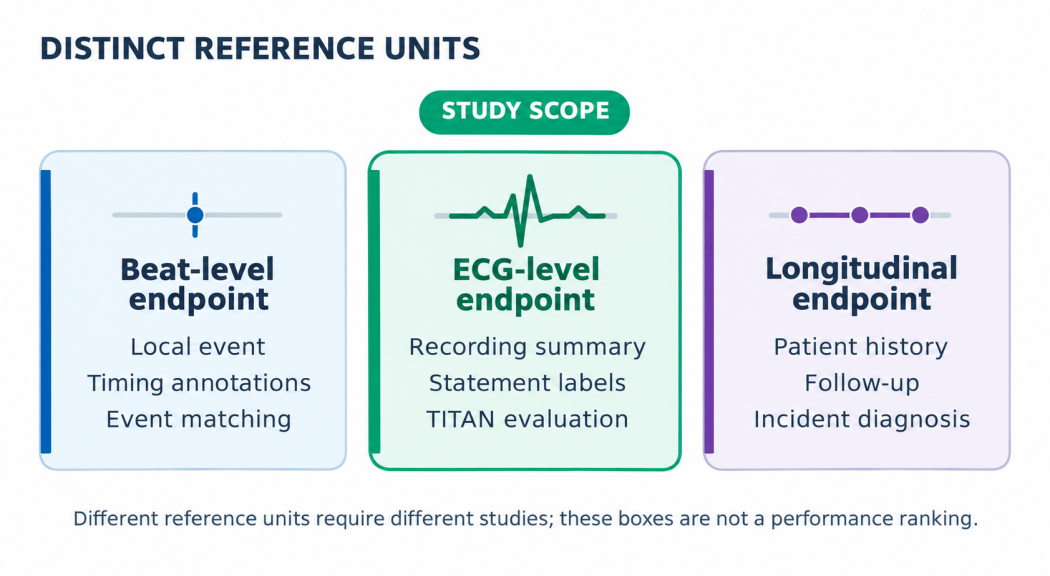


Figure 2. Conceptual comparison of beat, ECG and longitudinal reference units. The study evaluates ECG-level statements only.

The proposed architecture combines local waveform feature extraction with sequence modeling. Residual convolution, channel reweighting, and self-attention provide established methodological foundations for these components [9–11]; their combination must nevertheless be assessed within the specific ECG task and evaluation design. TITAN uses separately fitted rhythm and pathology backbones of the same architecture, each paired with its specialist. Primary6 (P6) reports six selected rhythm and conduction statements, whereas Primary4 (P4) evaluates four structural or ischemic ECG statement endpoints in separate binary panels. These profiles define the outputs evaluated in this study; they do not establish correct routing or diagnosis for ECGs outside those reporting boundaries.

1.1. State of the Art and Comparative Literature

Published studies illustrate the range of related tasks and evaluation designs. PTB-XL benchmarking compares approaches to ECG statement prediction [4], while Li et al. combine within-beat and recording-scale information in a dual-view CNN–Transformer [12]. Vranken et al. examine uncertainty estimation, and Herman et al. validate automated interpretation using expert diagnostic consensus as the reference standard [7,13]. Other approaches include feature extraction for heartbeat classification [14], channel-attention CNNs applied to segments labeled by beat category [15], and a multiscale hierarchical CNN evaluated on heartbeats in MIT-BIH and recordings in CCDD [16]. Differences in their targets and evaluation units require these contributions to be considered separately from the selected TITAN ECG and panel endpoints.

The comparison in Table 1 is purposive. Studies with multi-lead ECG inputs or related ECG-statement outputs provide the closest task context; heartbeat and clinical-workflow studies illustrate other methods and evaluation settings. Venue was not a selection criterion. Each published score belongs to its own label definitions, population, unit of analysis, and validation design, which prevents a direct ranking of the systems. The table provides context and does not constitute a systematic review or matched performance benchmark. NR identifies information that was not reported or was not comparable on the cited basis.

The objective of this study was to design a modular 12-lead ECG classifier, characterize the inputs and outputs of its specialists, and quantify accuracy, classwise discrimi-

Table 1. Selected ECG-AI studies across venues, compared by input, task and evaluation unit. DNN, deep neural network; SVM, support vector machine; KNN, k-nearest neighbors; NR, not reported on a comparable basis. DERMA and FrIFT retain the source method names. The present study is excluded from this prior-work table.

Study / venue	Scope	Architecture / parameter reporting	Outputs and endpoint role
Mastoi et al. (2022), Life	ECG heartbeat classification with DERMA fusion	DERMA and FrIFT; SVM or KNN	Heartbeat classification; methodological context for 12-lead rhythm-plus-pathology specialist reporting
Ribeiro et al. (2020)	Large 12-lead diagnostic-abnormality classifier	Residual DNN; comparable params NR	Six diagnostic abnormalities; direct 12-lead comparator with different label scope and no cascade contract
Zhu et al. (2020)	Multi-label rhythm and conduction diagnosis	DNN; comparable params NR	Multi-label rhythm/conduction outputs; direct ECG-AI comparator with different ontology and no pathology specialist
Strodthoff et al. (2021)	PTB-XL benchmark across ECG deep learning models	Varies by benchmark model	SCP-style multi-label outputs; benchmark reference rather than one matched specialist architecture
Vranken et al. (2021)	Uncertainty-aware 12-lead ECG analysis	DNN; comparable params NR	12-lead interpretation with uncertainty review; relevant to audit and uncertainty reporting
Herman et al. (2024)	Clinical validation of automated 12-lead interpretation	Six deep neural networks	Broad ECG statements; clinical validation context rather than transparent architecture-size comparator
Li et al. (2024)	ECG abnormality classification	Dual-view CNN–Transformer	Beat and recording-scale information; distinct targets
Elsheikhy et al. (2025), Sensors	Beat classification in MIT-BIH and INCART	CNN with channel attention	Segments labeled by five beat categories, not P6/P4
Zhou et al. (2025)	Multi-lead ECG classification with hierarchical features	Hierarchical CNN; parameter count reported	Heartbeats in MIT-BIH; recordings in CCDD; not an ECG-level clinical cascade

nation, and calibration in explicitly defined retrospective populations. Evaluation of the released reporting profiles is distinguished from later analyses of denominator sensitivity, morphology inputs, and selective prediction. The resulting estimates describe the populations and endpoints actually evaluated and do not measure a complete clinical diagnostic workflow. Independent external validation remains a separate, unfinished study.

2. Materials and Methods

2.1. Study Design and Evidence Boundary

This study designed and evaluated CNN–Transformer specialists on standardized ECG windows. The original release identifier `external_dev`, termed external-development in the submission, refers to archived selected development evaluations rather than a single untouched test set. V3H used its 2,193-ECG P8 holdout for development validation; the selected P6 profile comprises 1,643 ECGs, and epoch selection used the reported P6 population. P4 uses four selected binary panels, with 962 panel observations from 733 unique ECGs, not 962 distinct patients. Not every panel is evaluated for every ECG. Table 2 defines the profiles, and Table 3 separates reporting choices from clinical decisions.

Independent external evaluation is a separate, unfinished analysis. Its intended confirmatory design binds cohort eligibility, source and patient/signal overlap checks, label mappings, frozen checkpoint bytes, preprocessing, thresholds and analysis endpoints be-

fore outcome review. Cohorts inspected during model selection, threshold fitting or domain adaptation are not counted as independent. Any unresolved overlap or reference incompatibility must be reported rather than assumed absent. No performance estimate from the ongoing independent evaluation is presented. The completed within-CODE adaptation experiment is reported separately in Sections 2.8 and 3.10. P6 denotes Primary6, whose selected outputs are atrial fibrillation (AFIB), sinus bradycardia (SB), sinus tachycardia (STACH), right bundle branch block (RBBB), first-degree atrioventricular block (1AVB), and premature ventricular contractions (PVC). P4 denotes Primary4: anteroseptal myocardial infarction (ASMI), left ventricular hypertrophy (LVH), inferior myocardial infarction (IMI), and ischemic ST–T change (ISC_). P8 additionally reports normal sinus rhythm (NSR) and premature atrial contractions (PAC). These are ECG-associated reference statements rather than comprehensive clinical diagnoses.

Table 2. Present specialist evidence package reported in this manuscript.

Module	Role	Size/date	Output contract	Reported evidence
Backbone	Common 12-lead CNN–Transformer design	58,352,219	Separate fitted rhythm and pathology versions	Not used alone as a final clinical claim
Rhythm specialist	Selected Primary6 top-1 and binary membership	63,930,847	P6: AFIB, SB, STACH, RBBB, 1AVB, PVC; P8 adds NSR and PAC	95.19% accuracy; 80.09% macro-F1
Pathology specialist	Selected Primary4 binary panels	65,694,183	P4: ASMI, LVH, IMI, ISC_; other global outputs: ALMI, ILMI, LAE	80.23% mean accuracy; 79.35% macro-F1
Evidence boundary	Frozen manuscript package	2026-06-29	Tables, figures, thresholds, source summaries, record probabilities, and hashes	Supports selected-result calculation; not historical replay

Table 3. Key challenges in multi-source ECG-AI reporting and the corresponding response in the present manuscript.

Challenge	Why it matters	Study response	Where addressed
Diverse ECG sources	Sampling rates, acquisition devices, countries, and annotation policies differ across public databases.	Keep source identity, harmonize to a fixed 10 s 12-lead contract, and report source-stratified selected development results.	Data curation, source audit, and Results source-stratified plots
Flat label compression	Rhythm, conduction, ischemia, infarction, and hypertrophy endpoints can be mixed into one output surface.	Separate Primary6 rhythm, Primary4 pathology, and unreported outputs.	Class-contract tables and specialist figures
Hidden minority-class behavior	High pooled accuracy can hide low recall or low F1 for clinically relevant minority labels.	Use macro-F1, classwise panels, row-normalized confusion matrices, ROC/PR curves, and balanced one-vs.-rest pathology panels.	Results tables, confusion matrices, and PR/ROC figures
Uncertain predictions	Low-margin or high-entropy cases may appear as hard labels unless review signals are preserved.	Distinguish score uncertainty, top-1, and explicitly tested withholding.	Uncertainty, calibration, and Grad-CAM sections

2.1.1. Rationale for profile-specific reporting

NSR is a physiologic rhythm and may be frequent in unselected populations. Its omission from P6 defines a narrower classification problem; it does not correct selection bias or establish a method for separating normal from abnormal ECGs. Similarly, the four P4 panels test specified statements, not every infarct territory or all structural disease. The role table distinguishes evaluation eligibility from proposed handling outside the profile.

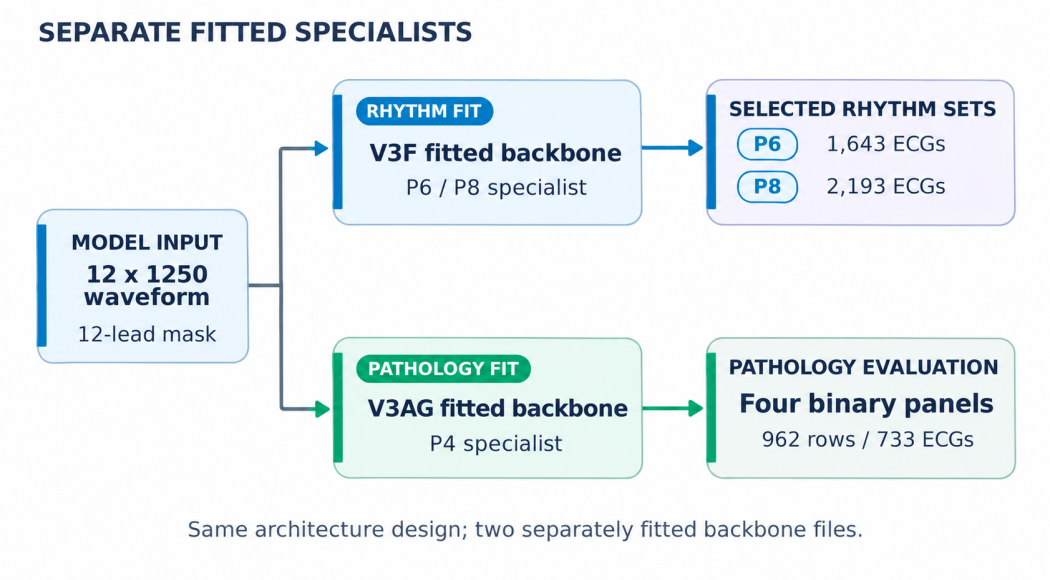


Figure 3. Specialist design. The P6/P8 and P4 evaluations use different fitted versions of the backbone design, V3F and V3AG, respectively.

2.2. Dataset Curation, Harmonization, and Population-Shift Control

Patient separation and waveform-duplicate checks matter because multiple ECGs or windows from one patient can share distinctive morphology. We distinguish source aliases and record-level exclusions from demonstrated patient/signal independence. The exact historical data loader and the train, calibration and validation rosters consumed by each job are not available for all checkpoints. Historical counts and verified retrospective preprocessing are therefore reported separately below.

Scalar quantities (f_s, T, $z_{i,c}$) are italicized, while column vectors ($\mathbf{x}_{i,\ell} \in \mathbb{R}^T$) and two-dimensional feature matrices ($\mathbf{X}_i \in \mathbb{R}^{L\times T}$) use bold lowercase and uppercase notation, respectively. The standard twelve-lead order is $\mathcal{L} = \{\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{aVR}, \mathrm{aVL}, \mathrm{aVF}, \mathrm{V1}, \mathrm{V2}, \mathrm{V3}, \mathrm{V4}, \mathrm{V5}, \mathrm{V6}\}$, with limb leads representing frontal-plane projections and precordial leads representing horizontal-plane views. An unstandardized recording from source institution i is represented as:

$$\mathbf{R}_i \in \mathbb{R}^{L_i \times T_i^{\mathrm{raw}}}, \qquad f_i^{\mathrm{raw}} > 0, \tag{1}$$

where $L_i \leq 12$ is the number of active recording leads, T_i^{raw} is the native discrete temporal length, and f_i^{raw} is the device-specific analog-to-digital sampling frequency.

Multi-center clinical datasets differ in acquisition hardware, analog filtering and sampling protocols. Before tensor input, the temporal resampling and lead-alignment operator $\mathcal{R}_{f_i^{\mathrm{raw}} \to f_s}$ standardizes their dimensions:

$$\mathbf{X}_i = \mathcal{R}_{f_i^{\mathrm{raw}} \to f_s}(\mathbf{R}_i, \mathcal{L}, 10\ \mathrm{s}), \qquad \mathbf{X}_i \in \mathbb{R}^{12\times 1250}, \quad f_s = 125\ \mathrm{Hz}. \tag{2}$$

The tensor contract uses 125 Hz and ten seconds. Resampling and named-lead alignment prepare the signal input; inference over the label ontology remains a separate step. The model receives a lead-validity mask separately from the waveform values. This representation does not guarantee preservation of every high-frequency diagnostic feature or reconstruction of absent leads.

Electrode contact impedance, patient respiration and amplifier calibration gains can introduce low-frequency baseline fluctuations and scale differences across multi-center recordings. Lead-wise z-score standardization reduces within-lead offset and scale variation independently within each temporal analysis window. It does not preserve absolute voltage or all cross-lead voltage ratios:

$$\tilde{\mathbf{x}}_{i,\ell} = \frac{\mathbf{x}_{i,\ell} - \mu_{i,\ell}\mathbf{1}}{\sigma_{i,\ell} + \epsilon}, \qquad \ell \in \mathcal{L}, \tag{3}$$

where $\mu_{i,\ell}$ and $\sigma_{i,\ell}$ are the sample mean and empirical standard deviation of lead ℓ within the active window, and $\epsilon = 10^{-8}$ provides numerical stability during near-isoelectric intervals. Normalization statistics are calculated within the local window of each tracing rather than across pooled partitions; this does not prevent patient or signal overlap.

For extended diagnostic recordings sampled into multiple overlapping or sequential sliding windows W_i, record-level pooling aggregates predictions before nonlinear activation. The uncalibrated window-level logit vectors $z_{i,w,c}$ for class c are averaged across all windows before the logistic sigmoid is applied:

$$z_{i,c} = \frac{1}{W_i} \sum_{w=1}^{W_i} z_{i,w,c}, \qquad p_{i,c} = \sigma(z_{i,c}), \tag{4}$$

This illustrates record-level pooling before activation. It prevents counting each window as a separate ECG in this score, but does not convert ECG counts into patient counts. The released profiles use their documented score and activation paths.

2.2.1. Historical development populations and partitions

Files, windows, ECGs, patients and binary evaluation opportunities are distinct units. V3C records 9,021 training and 924 validation windows and removal of 390 record matches in its development roster. Job 16768 records 7,485 enumerated files and 9,945 windows: 9,021 train and 924 validation. The V3F reports separately record 9,021 rhythm training and 144 rhythm validation windows, and 5,013 pathology training and 776 pathology validation windows. These task-specific counts cannot be summed to count patients. The separate V3M/V3AG record-based split lists 9,885 training and 1,132 validation windows. P4 calibration job 17296 records 1,492 window–panel observations. Its exact roster and separation from all prior development stages were not authenticated.

PTB-XL, Chapman–Shaoxing, Ningbo, CPSC2018, CPSC2018-Extra, Georgia and CODE test appear among the retained aliases; a mixed local grouping is not a separate institution. Appendix A reports recoverable P6 and P4 source counts. Chapman–Shaoxing contributes 1,514 of 1,643 P6 ECGs. The joint training and validation table by source, class, window and patient cannot be recovered. A conditional metadata reconstruction for V3M/V3AG flagged possible shared patients despite distinct record identifiers. It neither identifies the bytes consumed by historical jobs nor demonstrates contamination of PTB-XL fold 10.

2.2.2. ECG-associated reference statements

For retained Challenge-format rows, atrial fibrillation (AF) and normal sinus rhythm (NSR) correspond to SNOMED CT codes 164889003 and 426783006. CPSC, Georgia,

Chapman–Shaoxing and Ningbo rows use source-associated diagnosis fields, not a newly adjudicated reference. PTB-XL uses source Standard Communications Protocol for Computer-Assisted Electrocardiography (SCP-ECG) metadata, with mapped codes in its Challenge representation. CODE test uses supplied references; the mixed local group cannot be assumed to have a uniform annotation process. Source procedures differ and the archive does not authenticate all individual readers [5,6].

Coexisting eligible labels are kept in the accepted reference set. Top-1 is correct if its class belongs to that set. One record contains both AF and NSR. P8 retains both and P6 excludes NSR by profile definition, not by adjudication. The AF code and retained reference agree in all 1,643 P6 and 2,193 P8 rows. Each overlapping profile has 390 AF-positive rows (373 Chapman–Shaoxing, 13 mixed local, four PTB-XL Challenge); the counts cannot be added. Other aliases have no AF-positive reference in these selected records, which does not estimate parent prevalence.

P4 references are panel-specific SCP-derived statements. The recorded label agrees with the relevant SCP key in 962/962 panel observations. This is mapping concordance, not an independent ECG reading or clinical adjudication. AF recognition is ECG-associated statement prediction, not detection of first diagnosis or incident disease, which requires longitudinal evidence [17].

Figure 4 separates the reconstructed preprocessing steps from the provenance of the historical selected profiles and the unfinished independent assessment. Harmonizing waveform dimensions does not harmonize source annotation standards or demonstrate patient isolation.

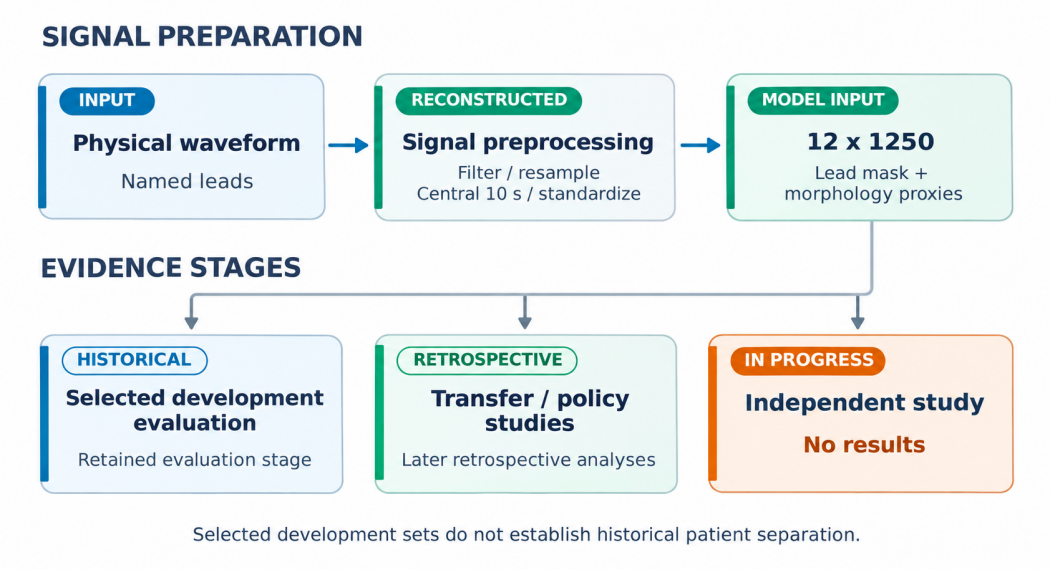


Figure 4. Signal preprocessing and distinct evidence stages. Selected development sets do not establish historical patient separation.

2.2.3. Source databases and reference compatibility

Table 4 distinguishes source families retained in historical development from those used in later analyses. A source name identifies an acquisition or repository family; an alias in a derived file does not necessarily identify a different hospital or an independent patient population. PTB-XL has both native SCP-coded and Challenge-format representations. A version derived at a different sampling rate also does not create a new cohort. The following descriptions address reference provenance and intended analytical use without authenticating all historical source partitions.

PTB-XL comprises ten-second 12-lead ECGs with diagnostic, rhythm and form statements, a hierarchy for diagnostic labels, and source patient identifiers [5]. Its published patient-based folds provide a contemporary partition convention, but a fold identifier in a recovered evaluation table does not establish which records every preceding training job consumed. The P4 evaluation uses selected positive and negative panels derived from SCP statements. These panels do not represent infarction, hypertrophy or ischemia prevalence. The four panels overlap at the ECG level, and a known target in one panel does not imply that all other pathology targets are known for that ECG.

Chapman–Shaoxing is a Chinese 12-lead arrhythmia database described by Zheng et al. [18]. It dominates the retained P6 sample. CPSC2018, CPSC2018-Extra, Georgia and Ningbo are retained as distinct source labels in the Challenge-associated files [6]. Their diagnoses are mapped to a shared output vocabulary while their original annotation procedures remain source-specific. CODE test is the public expert-reference test resource associated with the Brazilian 12-lead study of Ribeiro et al. [2]; the 19 selected P6 records from that resource have already contributed to the reported development evidence. They cannot subsequently serve as a newly untouched external cohort for this same performance claim.

SPH denotes *Shandong Provincial Hospital*. Its large multi-label database provides standardized diagnostic statements [19]. The completed retrospective experiment used 25,770 ECGs and 24,666 source patient identifiers. Its mapping supplies five P6, six P8 and three P4 compatible endpoints. A source Normal-only statement is not equivalent to a positive NSR rhythm annotation. Absence of a mapped diagnostic statement supports an absent-statement comparison within this mapping; it does not establish an independently adjudicated disease-negative control. This distinction matters particularly for rare pathology endpoints, because a large number of absent statements can strongly alter predictive values.

LUDB provides 200 ten-second 12-lead recordings with waveform annotations and diagnostic metadata [20,21]. The retrospective route study used all 200 ECGs. Explicit principal rhythm contrasts support some endpoints, whereas incidental findings may provide positives without reliable negative references. For example, LVH-positive and unknown LUDB records must not be combined into a conventional positive/negative P4 panel. PTB Diagnostic is a separate source containing 549 ECGs from 290 subjects [22]. The pathology transport analysis used named infarct territories as positive references and explicitly healthy controls; other records were retained as unknown for each territory comparison. Neither LUDB nor PTB Diagnostic was treated as a replacement for the missing parent population of the June specialist panels.

2.2.4. Partition accounting and exclusions

Table 5 records recoverable population counts without combining incompatible units. A file can contain several windows; a patient can contribute several files; and one ECG can contribute to more than one binary pathology panel. Consequently, the total number of training patients cannot be obtained by adding task-specific window counts. The source counts of the archived job 16768 sum to its 9,945 windows: CPSC2018, 3,394; CPSC2018-Extra, 45; Georgia, 659; INCART, 359; Ningbo, 22; PTB-XL, 5,420; PTB-XL Challenge 2020, three; extra 3AVB, 29; and Ningbo 3AVB, 14. These are source-attributed windows for that run, not a release-wide source-by-class training census.

The V3C record-match removal is documented, but it does not certify removal of every patient or signal duplicate from all subsequent stages. The recovered V3M/V3AG record split supports a conditional reconstruction that flags possible shared patients. It does not identify the exact bytes consumed by historical jobs or establish contamination

Table 4. Source families, reference types and analytical roles. Counts are study-specific observations, not a reconstructed joint source-by-partition roster.

Source / origin	Reference basis	Use in this study	Boundary
PTB-XL / Germany	SCP diagnostic, rhythm and form statements	P4 selected panels; historical development; later folds 1–9 bridge	Native and Challenge aliases can overlap
Chapman–Shaoxing / China	Source rhythm diagnoses	1,514 selected P6 ECGs	Dominant retained source; historical patient isolation unverified
CPSC2018, Extra; Ningbo / China	Challenge-associated diagnostic codes	Historical development and selected P6 aliases	Source-specific annotation; small selected strata
Georgia / USA	Challenge-associated diagnostic codes	Historical development and six selected P6 ECGs	Small retained evaluation stratum
CODE test / Brazil	Supplied expert-reference labels	19 selected P6 ECGs	Already inspected; no new independent-test claim
Mixed local group	Retained heterogeneous reference fields	63 selected P6 ECGs	Not one institution or a uniform adjudication protocol
SPH / China	Standardized multi-label statements	Full 25,770-ECG retrospective analysis	Missing 1AVB, NSR and ISC_ mappings
LUDB / Russia	Principal rhythm and diagnostic metadata	200-ECG route and reconstruction studies	Unknown references retained as unknown
PTB Diagnostic / Germany	Subject diagnosis, named territory, healthy controls	549-ECG pathology transport analysis	Endpoint-specific known sets; unknowns excluded only from endpoint metrics

Table 5. Recoverable historical development and selected-evaluation counts. NR means not recovered with sufficient job-level binding.

Stage	Training	Validation / calibration	Statistical unit and limit
V3C / job 16768	9,021	924 validation	Windows; 7,485 enumerated files; 390 record matches removed in roster construction
V3F / job 16781, rhythm	9,021	144 validation	Task-specific windows; eight reported epochs
V3F / job 16781, pathology	5,013	776 validation	Task-specific windows, not distinct patients
V3M / V3AG split	9,885	1,132 validation	Record-based split; possible patient overlap in conditional reconstruction
P4 calibration / job 17296	NR	1,492	Window–panel observations; exact consumed roster NR
Released P6 selected profile	Not inferred	1,643 evaluation ECGs	Development and model selection
Released P8 profile	Not inferred	2,193 evaluation ECGs	Development validation; overlaps P6
Released P4 panels	Not inferred	962 panel observations	733 distinct ECGs; not 962 patients

of the official PTB-XL fold-10 test partition. Conversely, a bounded signature search without a demonstrated collision does not prove independence of all historical data. Missing consumed rosters and incomplete checkpoint ancestry prevent a definitive historical independence statement.

Window preparation follows the fixed tensor contract; technically invalid or incompatible references are tracked separately from disease-negative labels. The selected evaluation files do not retain a complete preselection population with every exclusion reason. Therefore, a June all-input enrollment diagram, source-by-class-by-patient partition table or natural-prevalence clinical predictive value cannot be reconstructed from the selected records alone. Later all-input studies preserve their own enrolled denominators and technical failures. Their eligibility accounting is reported with those studies rather than retroactively attached to the original June analysis.

We classify historical partition evidence into three states. *Recorded* denotes surviving job summaries, loss/launcher excerpts or archived selected prediction rows. These support the quantities they contain, including windows or ECG–label observations, but do not identify the inputs actually consumed by every trainer. *Reconstructed* denotes later joins of source metadata, aliases, folds and record identifiers, which permit conditional overlap checks but cannot certify the missing job inputs. *Uncertified* denotes the unrecovered joint source-by-class-by-partition patient roster, job-bound label masks, input checkpoint ancestry and complete historical patient/signal separation. Uncertified cells remain unavailable, rather than zero or estimated. The contemporary within-CODE audit has its own sealed split evidence (Section 2.8) and does not repair this historical gap. Supplementary file `PARTITION_EVIDENCE_STATUS.csv` records these distinctions and their source files.

2.2.5. Independent external evaluation in progress

The purpose of the ongoing independent study is to test frozen ECG-statement prediction beyond the inspected development analyses. Independence is a property of the complete development and evaluation process, not of a database name. A candidate cohort can be considered confirmatory only after its admissible labels, patient and signal overlap checks, technical eligibility and prior inspection history have been documented. Already inspected SPH data, development-fold PTB-XL data, selected CODE test records and data used for fitting residual heads or adapting thresholds do not qualify automatically.

The completed retrospective analyses establish executable preprocessing and scoring contracts for specified checkpoints, but do not by themselves establish an admissible independent cohort. Before any independent performance estimate is incorporated, the analysis must bind the cohort roster and source version, label mapping and unknown-reference rules, checkpoint filenames and SHA-256 digests, inference configuration, score aggregation, membership cutoffs and statistical plan. Both source identifiers and waveform-duplicate checks are required because aliases or repeat acquisitions can evade record-name matching. Unresolved historical ancestry must remain a stated limitation even if the new cohort passes contemporary overlap checks.

The intended evaluation reports technical scoring coverage on all eligible ECGs, class-specific known reference counts, sensitivity, specificity, predictive values, F1, discrimination and calibration where both reference polarities support them. Patient or subject/signal components should define resampling units when repeated recordings are present. Any abstention policy requires its own frozen calibration cohort and a ledger separating emitted decisions, withheld known positives and technical failures. No independent-study sample size, completed inference count or performance result is as-

serted here. The current manuscript reports only the completed retrospective analyses and leaves independent external evaluation open.

2.3. Architecture and Specialist Branches

Convolution and attention operate at different temporal scales. The former builds local waveform features, while the latter integrates information across the ten-second window. The description below summarizes the model design; the observed tensor widths and branch connectivity are reported separately.

Let $\mathbf{X}_i \in \mathbb{R}^{12\times 1250}$ denote the harmonized multi-lead input matrix. At residual stage b, the convolutional feature extractor maps the intermediate tensor $\mathbf{H}_b \in \mathbb{R}^{C_b \times T_b}$ to $\mathbf{H}_{b+1} \in \mathbb{R}^{C_{b+1}\times T_{b+1}}$ as follows:

$$\mathbf{H}_{b+1} = \phi(F_b(\mathbf{H}_b; \boldsymbol{\theta}_b) + P_b(\mathbf{H}_b)), \tag{5}$$

where $F_b(\cdot; \boldsymbol{\theta}_b)$ combines sequential one-dimensional convolutions, batch normalization layers and dropout regularization, with weight tensors $\boldsymbol{\theta}_b$. The shortcut $P_b(\cdot)$ is an identity or linear 1×1 projection designed to support gradient propagation through deep layers, and $\phi(\cdot)$ denotes the Rectified Linear Unit (ReLU) activation function [9]. This convolutional stem learns local temporal features from the prepared multi-lead waveforms; the experiment does not validate individual filters as electrophysiological measurements.

Standard 12-lead acquisitions project cardiac dipoles onto distinct anatomical planes, so diagnostic information varies across channels. For example, anterior myocardial necrosis alters precordial leads V1–V4, whereas inferior-territory changes may involve leads II, III, and aVF. A Squeeze-and-Excitation channel attention mechanism is designed to prioritize informative projections and reduce the influence of noisy channels. It computes a recalibration vector $\mathbf{s} \in (0,1)^C$ from the post-residual feature representation $\mathbf{U} \in \mathbb{R}^{C\times T'}$ [10]:

$$\mathbf{s} = \sigma(\mathbf{W}_2\,\mathrm{SiLU}(\mathbf{W}_1\,\mathrm{GAP}(\mathbf{U}))), \qquad \widehat{\mathbf{U}} = \mathbf{s} \odot \mathbf{U}, \tag{6}$$

where $\mathrm{GAP}(\cdot)$ averages activations across the temporal dimension to produce channel-wise summaries. The linear transformation matrices $\mathbf{W}_1$ and $\mathbf{W}_2$ form an excitation bottleneck with smooth Sigmoid Linear Unit (SiLU) activation, and $\sigma(\cdot)$ is the logistic sigmoid gate that scales the channel modulation vector $\mathbf{s}$. The element-wise Hadamard product $\odot$ gives the recalibrated tensor $\widehat{\mathbf{U}}$ by reweighting learned feature channels. These feature channels must not be equated one-to-one with physical leads.

A multi-head scaled dot-product self-attention Transformer encoder processes the recalibrated representations to capture non-local patterns across multiple cardiac cycles [11]. It models contextual interactions across arbitrary temporal distances within the tracing, with attention defined as:

$$\mathrm{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{7}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the query, key, and value matrices projected from the latent sequence. The denominator $\sqrt{d_k}$ stabilizes variance to limit gradient saturation at large inner products. The Transformer uses 10 parallel self-attention heads with latent embedding dimension $d_{\text{model}} = 640$ and a position-wise feed-forward expansion layer of dimension $d_{\text{ff}} = 2560$.

The complete computational allocation and topological routing mechanisms are documented in Table 6 and Table 7. Each loaded CNN–Transformer backbone object contains 58,352,219 parameters including its global heads. The rhythm package and pathology package comprise 63,930,847 and 65,694,183 parameters, respectively, including their sep-

arate fitted backbones. Reconstructed tensor interconnectivity and reporting roles are illustrated in Figure 5 and Figure 6.

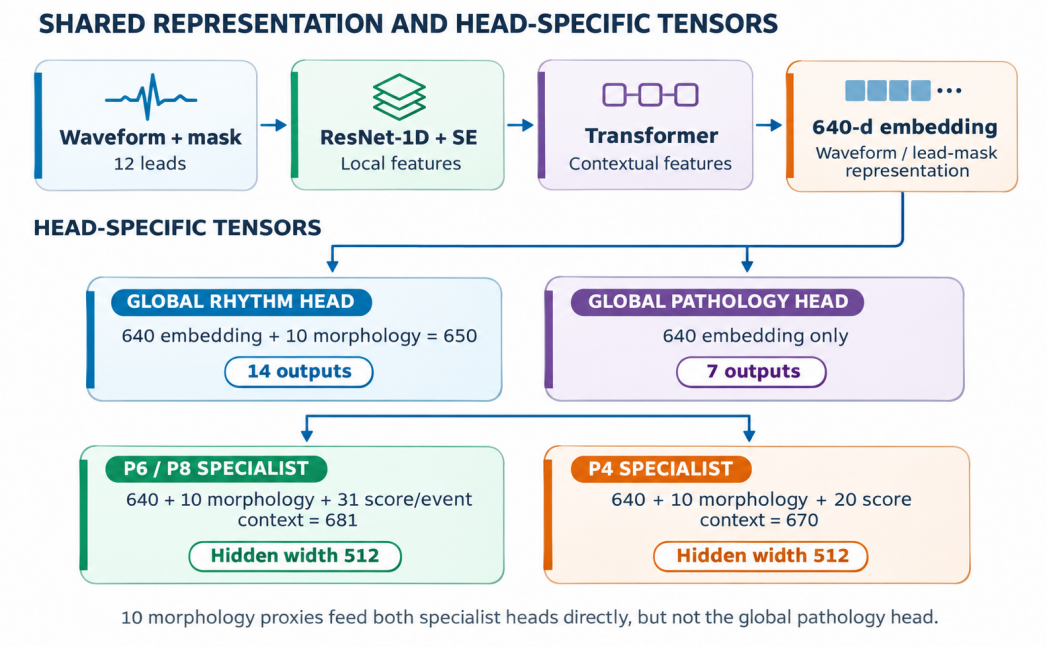


Figure 5. Instrumented CNN–Transformer and specialist data flow. Arrows denote tensors passed in the reconstructed runtime. The ten morphology proxies feed both specialists directly; the global pathology head does not consume them directly.

Table 6. Architecture and parameter-count summary.

Module	Description	Parameters	Scope
Input	12-lead ECG	-	12 x 1250 at 125 Hz
Backbone design	ResNet-1D + Transformer + global heads	58,352,219	Separate multi-head representation
Rhythm package	Backbone + classwise rhythm specialist	63,930,847	Primary6 operational branch
Pathology package	Backbone + 512-unit pathology specialist	65,694,183	Primary4 operational branch

Note: Package totals include the corresponding fitted backbone and its global heads plus the attached specialist. The two package totals must not be added and described as one shared forward pass.

Table 7. Primary, secondary-candidate and auxiliary class contracts.

Task	Role	Classes
Rhythm	Primary	AFIB, SB, STACH, RBBB, 1AVB, PVC
Rhythm	Secondary candidate	NSR, PAC, Flutter, Paced
Rhythm	Auxiliary global	LBBB, 2AVB, 3AVB, LQTS
Pathology	Primary	ASMI, LVH, IMI, ISC_
Pathology	Secondary candidate	ALMI, ILMI
Pathology	Auxiliary global	LAE

2.3.1. Output definitions and direct supervision

Global rhythm and pathology heads expose 14 and seven outputs. P6 reports AFIB, sinus bradycardia (SB), sinus tachycardia (STACH), right bundle branch block (RBBB), first-degree atrioventricular block (1AVB), and premature ventricular contractions (PVC).

P8 adds NSR and premature atrial contractions (PAC). P4 reports anteroseptal myocardial infarction (ASMI), left ventricular hypertrophy (LVH), inferior myocardial infarction (IMI), and ischemic ST–T change (ISC_). V3F and V3AG are separate released backbone files. Shared design does not imply identical fitted weights.

Recovered V3F code indexes P8 in its rhythm loss (0.70 focal binary cross-entropy plus 0.30 categorical cross-entropy) and P4 in its pathology focal binary loss. Flutter, paced rhythm, left bundle branch block (LBBB), second- and third-degree atrioventricular block (2AVB and 3AVB), and long QT syndrome (LQTS) have no direct term in that V3F objective. Separate V3AG development code uses a different rhythm subset including Flutter and paced rhythm. Anterolateral and inferolateral myocardial infarction (ALMI and ILMI) and left atrial enlargement (LAE) are outside the recovered V3F direct pathology objective. These code excerpts do not reconstruct earlier supervision or bind every runtime mask to released weights. Auxiliary metrics from the separate 41M experiments cannot be attributed to V3F or V3AG. LQTS is not interchangeable with a prolonged-QT statement, nor is LAE automatically equivalent to left atrial overload (LAO).

In Table 7, *Role* identifies a reporting or output contract. Primary outputs enter that profile's metrics; secondary candidates are outside the report and were proposed for separate handling; auxiliary endpoints are global outputs outside the specialist evaluation. These categories are not dispatch decisions. NSR is normal rhythm, excluded from P6 but included in P8. A residual Other group requires a specified ontology and reference; NSR should not be called Other arrhythmia.

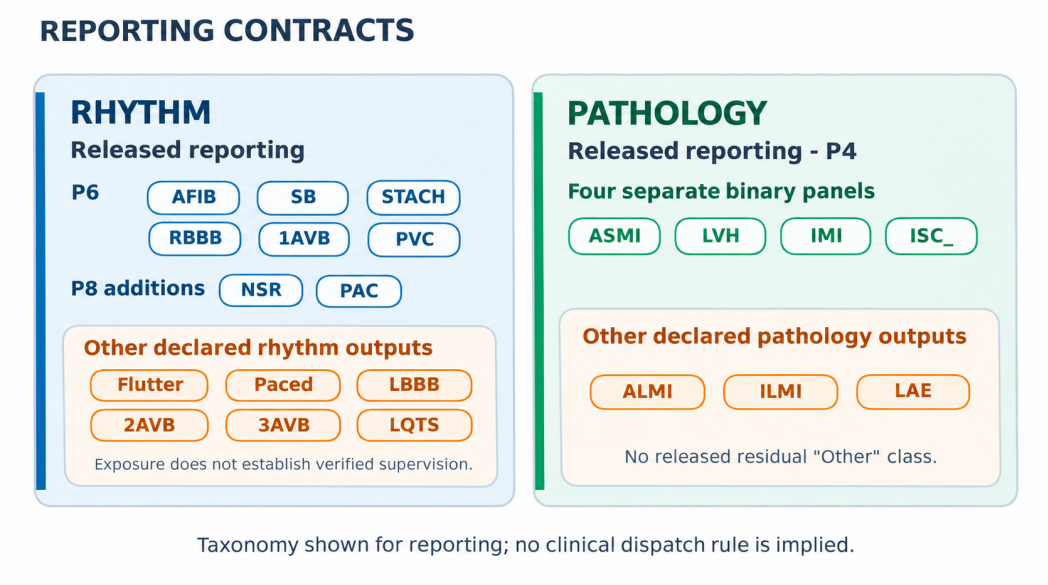


Figure 6. Reporting and output contracts. Groupings do not describe a clinical dispatch rule and no arrow implies movement between groups.

Supplementary Table S1 distinguishes global-logit availability, recovered stage-specific supervision, published specialist evaluation and later experimental evidence for all eleven out-of-profile or auxiliary endpoints. A documented loss index is not a recovered patient-level training roster. An experimental head or newly selected threshold does not become a published specialist decision.

2.3.2. Reconstructed morphology inputs and tensor connections

The reconstruction started with physical millivolt waveforms. It applied a causal third-order Butterworth filter (0.5–45 Hz on nonconstant leads), polyphase resampling from 500 to 125 Hz, extraction of the central ten seconds with padding when required,

and per-lead standardization. The 12×1250 waveform has a 12-element lead mask. This contract was instrumented on LUDB, not authenticated as the exact historical loader. Inter-peak RR intervals supply the timing proxies, including the root mean square of successive differences (RMSSD). Coefficient of variation and spectral ratios are dimensionless.

The extractor uses lead II if nonflat, otherwise the highest-variance lead. Median-centered magnitude peaks meet or exceed the larger of the 88th percentile and mean plus 0.8 standard deviations and are separated by at least 0.28 s. For peak sample indices p_i, inter-peak intervals $RR_i = (p_{i+1} - p_i)/f_s > 10^{-6}$ are in seconds, with $f_s = 125$ Hz in the reconstructed path. For lead ℓ, let $P_{\ell k} = |\,\mathrm{rFFT}(x_\ell)_k|^2$ and $\bar{P}_k$ be its across-lead mean. In the energy ratios, $E_{a:b} = \sum_{k:a\le f_k\le b} \bar{P}_k$ and $E_{\mathrm{tot}} = \sum_k \bar{P}_k$; band endpoints are inclusive. Limb asymmetry uses lead-specific sums $E_\ell = \sum_{k:5\le f_k\le 18} P_{\ell k}$. These are unnormalized squared Fourier magnitudes of prepared samples, not physical power measurements. A total at or below 10^{-12} produces a zero band ratio. Values are divided by the listed scales and clipped to $[0, 1]$ (signed asymmetry $[-1, 1]$); nonfinite values become zero.

Table 8. Ten morphology descriptors. Width and pre-peak lag are proxies, not clinical intervals.

Descriptor	Computation and unit	Scale
Median heart rate	$60/\mathrm{median}(RR)$, beats/min	220
RR variation	Population $\mathrm{SD}(RR)/\mathrm{mean}(RR)$	1
RMSSD	$10^3\sqrt{\mathrm{mean}[(RR_{i+1} - RR_i)^2]}$, ms	500
QRS-width proxy	Median inclusive span above max(35% segment maximum, 70th percentile) within ± 0.18 s of a peak, ms	220
P-to-R proxy	Median lag to maximal pre-peak deflection 0.08–0.28 s before peak; retain lags 60–320 ms	320
QRS-band energy	$E_{5:18}/E_{\mathrm{tot}}$	1
Low-frequency energy	$E_{0:0.7}/E_{\mathrm{tot}}$	1
High-frequency energy	$E_{35:\mathrm{Nyq}}/E_{\mathrm{tot}}$	1
Cross-lead correlation	Mean finite absolute off-diagonal Pearson correlation over leads with $\mathrm{SD} > 10^{-6}$	1
Limb-lead asymmetry	$(E_I - E_{aVF})/\max(E_I + E_{aVF}, 10^{-12})$ in 5–18 Hz	1

The QRS-width proxy uses the first and last threshold-active sample indices a_i, b_i, including intervening gaps: $1000(b_i - a_i + 1)/f_s$ ms. The pre-peak search uses the half-open sample interval from $p_i - \mathrm{round}(0.28 f_s)$ to $p_i - \mathrm{round}(0.08 f_s)$, bounded by the window. The table's divisors use the corresponding raw units, so all ten emitted features are dimensionless after scaling. Nonfinite input samples are replaced by zero before extraction; undefined derived descriptors become zero during normalization. Supplementary file `MORPHOLOGY_FORMULA_CODE_AUDIT.csv` maps every descriptor to the recovered function, unit, divisor and boundary rule. The extractor matches the source bound to the reconstructed runtime, rather than certifying the unavailable historical loader.

The global rhythm head receives a 640-dimensional waveform/mask embedding and ten descriptors (width 650). The global pathology head receives the embedding without direct morphology. P6/P8 (681 inputs) and P4 (670 inputs) both receive morphology at indices 640–649 alongside task-specific score or event inputs. P4 also receives rhythm scores. Stored rhythm head width 512 supersedes stale metadata declaring 192; P4 hidden width is also 512. The 58,352,219-parameter backbone count includes the global heads, and does not describe the separate 41M models.

A flat-signal fixture produced spurious plateau peaks and nonzero descriptors. The proxies are not clinically validated. Job 33796 instrumented both tensor paths on 16 LUDB ECGs and exactly repeated a reconstructed fixed-batch forward. This is not historical waveform replay: a separate strict P4 comparison exceeded the score tolerance for 836/962 observations even though threshold decisions agreed in that comparison.

2.3.3. Separate retrospective analyses

The Shandong Provincial Hospital (SPH) analysis applied released profiles to all 25,770 metadata ECGs (24,666 source patient identifiers) before forming nested profile-compatible subsets from the same scores. Absence of a compatible statement is negative for the mapped endpoint, not adjudicated absence of disease. The primary SPH mapping does not supply 1AVB, NSR or ISC_ references. A PTB-XL folds 1–9 development cache supplied separate 1AVB and ISC_ statements; 6,205 of 19,601 metadata ECGs were technically scored. LUDB and PTB Diagnostic are separate sources and cannot replace SPH or the missing June population.

Other experiments fitted new development heads. The rhythm residual covers mapped out-of-profile statements; normal rhythm and unknown references remain distinct. The pathology residual is a bounded PTB-XL ALMI/ILMI/LAO-or-LAE union. A morphology ablation fitted equal-capacity heads with fixed backbone representations and five matched seeds per arm. Simultaneous intervals were assessed against a predefined ± 2 percentage-point relevance margin. These experiments do not constitute independent validation or full-network retraining, and they do not change the released models.

2.3.4. Specialization, loss masks and output semantics

A global output index, a supervised loss term, a specialist output and an evaluated endpoint are distinct. An available logit for a named diagnosis shows that the implementation can emit that score; it does not show that the output received direct training support in every stage. Recovered V3F loss indexing supports the stated P8 and P4 objectives, but the supervision used in earlier or different development families cannot be inferred from the released architecture alone. In particular, the separate historical 41M experiments do not document direct supervision for every output of the released 58.35M backbones.

Primary, secondary-candidate and auxiliary are reporting roles in Table 7. They do not identify mutually exclusive patient groups. A conduction statement such as RBBB may coexist with a rate or rhythm statement, and a pathology reference can coexist with any rhythm. A categorical top-1 view compresses that set into one reported class, whereas thresholded membership can retain several classes or none. P6 and P8 therefore require both an accepted-reference-set definition and a separate membership policy. Calling NSR an arrhythmia or treating it as a residual abnormal class would change the task. A residual Other endpoint must instead define which out-of-profile statements count as positive and which records have genuinely known negative references.

ALMI and ILMI identify infarct-territory statements distinct from ASMI and IMI. Their availability as outputs does not make them evaluated P4 targets. LAE and LAO may be combined only in an explicitly bounded experimental reference, not relabeled as interchangeable clinical diagnoses. Similarly, LQTS is a syndrome label and cannot be validated by automatically substituting a prolonged-QT observation. The later residual experiments retain their actual target unions, eligible denominators and fitted heads. They do not create a newly validated released Other class.

2.3.5. Signal scaling and morphology limitations

Per-lead standardization reduces input dependence on recording gain but removes absolute amplitude and changes cross-lead amplitude relationships. The morphology power ratios and correlation proxies use this prepared representation and cannot recover the discarded physical millivolt scale. This limitation matters for LVH and other voltage-dependent findings, and motivates testing a future model that retains verified physical amplitudes alongside normalized waveforms. The present results do not isolate normal-

ization as the cause of LVH error, because no controlled amplitude-preserving ablation was performed.

The peak detector and pre-peak lag estimator are heuristic calculations. A maximum deflection before a detected peak need not be a P wave, and the thresholded width need not correspond to expert QRS onset and offset. Filtering, resampling, sampling phase, lead choice and artifact can all change the derived vector. A lead-validity mask describes the input contract but does not certify adequate signal quality. The known flat-signal failure is therefore reported as a methodological limitation rather than silently repaired before replay. Prospective use would require explicit quality criteria, interval-reference validation and failure handling evaluated on representative recordings.

Classical QRS detection provides a signal-processing context for automated peak localization [23]. The present extractor is defined by its own recovered thresholds and timing rules in Table 8. No timing-annotation study was performed to establish its peak or interval accuracy. Peak detection, interval measurement and ECG-statement classification must therefore remain separate claims.

The reconstructed runtime audit repeats a forward pass on the same prepared batch to test deterministic behavior in one numerical configuration. Reproducing the June calculation would also require the unavailable historical waveform windows, library versions, kernel choices, import paths and masks. Threshold agreement must therefore be interpreted alongside sensitivity to preprocessing and device arithmetic. Stable threshold decisions on one matched panel can coexist with non-equivalent scores and with changed decisions on other ECGs.

2.4. Loss Function, Optuna Search, Fine-Tuning, and Calibration

Class imbalance motivated a focal binary term [8]. For a known binary target y_{ic}, let $p_{t,ic} = p_{ic}$ when $y_{ic} = 1$ and $p_{t,ic} = 1 - p_{ic}$ otherwise. With eligibility mask m_{ic}, the focal term is

$$\mathcal{L}_{\text{focal}} = -\frac{\sum_{ic} m_{ic}\alpha_{ic}(1 - p_{t,ic})^{\gamma}\log(p_{t,ic} + \epsilon)}{\sum_{ic} m_{ic}}. \tag{8}$$

For stages enabling rhythm, pathology and clinical-axis objectives, the original multi-task design is expressed as

$$\mathcal{L}_{\text{multi}} = \mathcal{L}_{\text{rhythm}} + \lambda_{\text{path}}\mathcal{L}_{\text{pathology}} + \lambda_{\text{axis}}\mathcal{L}_{\text{axis}}. \tag{9}$$

Each stage must identify the active terms and masks. This design equation does not authenticate every historical coefficient or establish useful clinical-axis performance. Optimizer weight decay is reported as a configured parameter; it is not assumed to be a separately measured loss term.

Here α_{ic} denotes a class/target weight when used. Masked or unrecognized labels do not contribute to the corresponding term. The categorical component uses $-\sum_c q_{ic}\log\text{softmax}(z_i)_c$ for target vector q_i. The archived 50-epoch development run records auxiliary clinical-axis losses; it does not show that the released specialist decision has a validated auxiliary function. Exact historical configuration and label support are not available for all prior stages.

The public V3C selection summary describes a Tree-structured Parzen Estimator (TPE) sampler (summary seed 197813, multivariate mode, ten startup trials), with 12 planned trials and nine retained completed candidates, distinct from V4 and the 41M screening series [24]. The recorded epoch composite was

$$J_e = 0.6F1_{\text{P8},e} + 0.4F1_{\text{P4},e} - 0.05M_{\text{P8},e} - 0.05M_{\text{P4},e}, \tag{10}$$

where M denotes missing target classes or labels, not mini-batch variance. Candidate ranking uses the mean of the last three composites minus 0.25 times their population standard deviation. Trial 6 led the ranking (0.624512); trial 8 had the highest single-epoch peak (0.650940). The selection JSON and archived effective configuration agree on the parameters in Table 9. Job 16768 ran 50 epochs; job 16781 subsequently declared the intermediate checkpoint from that run as the initialization for V3F. The V3F report records eight epochs. The historical V3F output digest agrees with the published backbone. The original Optuna database, intermediate checkpoint bytes and job-level input/output digest binding remain unavailable. Neither scheduler completion nor the output digest proves the complete byte-level lineage.

Table 9. Recorded V3C trial 6 hyperparameters used in the archived job 16768 configuration.

Parameter	Selected value
base_lr	6.5005e-06
max_lr	9.7576e-05
weight_decay	7.9107e-06
gamma	1.5254
label_smoothing [25]	0.018121
batch_size	16
mixup_alpha	0.033822
lambda_pathology	0.24551

2.4.1. Thresholds, top-1 and empty prediction sets

For a binary statement, the thresholded decision is

$$\hat{y}_{ic} = \mathbb{1}[p_{ic} \geq \tau_c]. \tag{11}$$

For rhythm, the released top-1 output is distinct from the membership set. P6 retains the top-1 label on every selected ECG even when the binary set is empty; empty membership remains all-negative in P6 binary metrics. P8 fills an empty predicted set with the recorded top-1 class. The top-1 correctness criterion is membership in the accepted reference set. P4 remains a binary panel evaluation: below threshold is negative and remains in the panel denominator. Thresholds are the released values in Tables 13, 14 and A9. None of these cases is automatically a clinical abstention.

2.5. Uncertainty and selective prediction

Predictive entropy $H(p) = -\sum_c p_c \log p_c$ and the difference between the two highest probabilities can describe uncertainty for a categorical score. With T stochastic passes, the difference between entropy of the mean probability and mean entropy is a model-dependent diagnostic; it is not proof of out-of-distribution status. The original June report has no authenticated decision ledger connecting these statistics to actual withholding, secondary inference or expert review. Historical quarantine rates and pre/post accuracy are therefore not reported.

To preserve the original uncertainty formulation, let T_s denote a specified number of stochastic forward passes. Their mean categorical distribution is

$$\bar{p}(y = c \mid x) = \frac{1}{T_s} \sum_{t=1}^{T_s} p_{\theta_t}(y = c \mid x). \tag{12}$$

Its entropy and top-two margin are

$$H(\bar{p}) = -\sum_c \bar{p}_c \log \bar{p}_c, \qquad m = \bar{p}_{(1)} - \bar{p}_{(2)}. \tag{13}$$

The corresponding entropy-difference diagnostic is

$$I[y, \theta \mid x] = H(\bar{p}) - \frac{1}{T_s} \sum_{t=1}^{T_s} H(p_{\theta_t}(y \mid x)). \tag{14}$$

These equations define model-based summaries. They do not establish that the historical evaluation executed a particular number of passes, fitted an out-of-distribution detector or sent any patient to clinical review. Binary membership probabilities require their own uncertainty interpretation; they are not automatically one normalized categorical distribution. The later policy below uses threshold distance rather than these entropy summaries.

In the later score-based P8/P4 experiment, profile confidence was the minimum absolute distance between class scores and the respective released cutoffs. The profile cutoffs (P8: 0.037162775; P4: 0.014417143) were fitted for target coverage on 12,901 SPH calibration ECGs without reference labels. The frozen joint rule emits both decisions only if both profile cutoffs are met. Failure to produce scores is distinct from abstention. With N_s scored ECGs and N_a accepted ECGs, coverage is N_a/N_s and the scored abstention rate is $(N_s - N_a)/N_s$. Coverage relative to all enrolled ECGs is reported separately when some cannot be scored. Pre-policy risk is FP plus FN divided by known binary opportunities over all scored ECGs; post-policy risk uses only accepted opportunities. Withheld positives are separate from emitted false negatives. The full-denominator cost counts emitted errors plus λ times withheld known opportunities. A separate LUDB experiment runs actual P6 fallback only on withheld ECGs, with frozen P6 membership thresholds, no top-1 override and no clinician review.

2.6. Grad-CAM Explainability Contract

Post hoc saliency can support inspection of model sensitivity, although it does not establish physiological causality. The implementation provides class-discriminative one-dimensional Gradient-weighted Class Activation Mapping (1D Grad-CAM) [26] with a strict audit contract. For a target diagnostic class c, let y^c be its pre-activation score and $A_k(t)$ the intermediate temporal activation tensor in convolutional feature channel k. The spatial-temporal importance weights α_k^c are computed by averaging gradients across the temporal length T':

$$\alpha_k^c = \frac{1}{T'} \sum_{t=1}^{T'} \frac{\partial y^c}{\partial A_k(t)}. \tag{15}$$

The resulting channel-weighted linear combination passes through a Rectified Linear Unit activation to retain strictly positive contributions toward the target class, producing the continuous temporal saliency profile:

$$L^c_{\text{GradCAM}}(t) = \text{ReLU}\left(\sum_k \alpha_k^c A_k(t) \right). \tag{16}$$

The temporal heatmap indicates local gradient-weighted activation, not validated identification of an electrophysiological complex. This defines an explainability method and recommended metadata, not a validated explanation of decision causality or an inventory of all historical artifacts.

2.7. Evaluation Metrics and Confidence Intervals

For each known binary opportunity, TP, TN, FP and FN count true positives, true negatives, false positives and false negatives. With $N = TP + TN + FP + FN$,

$$\text{Acc}_{\text{bin}} = (TP + TN)/N, \quad \text{PPV} = \text{Precision} = TP/(TP + FP), \tag{17}$$

$$\text{Sensitivity} = \text{Recall} = TP/(TP + FN), \quad \text{Specificity} = TN/(TN + FP), \tag{18}$$

$$\text{NPV} = TN/(TN + FN), \quad F1 = 2TP/(2TP + FP + FN). \tag{19}$$

Undefined denominators are not treated as clinical zeros. With C class endpoints, MacroF1 $= C^{-1}\sum_c F1_c$; mean binary accuracy is $C^{-1}\sum_c \text{Acc}_{\text{bin},c}$. Top-1 accuracy is $n^{-1}\sum_i \mathbb{1}[\hat{c}_i \in S_i]$ for the accepted reference set S_i and one emitted top-1 class $\hat{c}_i$. P6 top-1 and mean binary F1 use different decision units; P4 mean binary accuracy is not top-1 or pooled panel accuracy.

With TPR and FPR denoting the true- and false-positive rates, AUROC $= \int_0^1 \text{TPR}\, d\text{FPR}$ for the binary ranking curve [27]. Average precision (AP) is $\sum_k (R_k - R_{k-1})P_k$ at successive distinct score thresholds, not trapezoidal PR area. Brier score is $n^{-1}\sum_i (p_i - y_i)^2$. For ten equally spaced probability bins, binary expected calibration error (ECE) is $\sum_b (n_b/n)|\bar{p}_b - \bar{y}_b|$ over nonempty bins. The vertical axis of the reliability plots is $\bar{y}_b$, the fraction of reference-positive observations in bin b.

The Wilson 95% interval for $k/n = \hat{p}$ is

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}, \quad z = 1.96. \tag{20}$$

Selected-source Wilson intervals use ECGs as observations [28]. Classwise F1 bars use the 2.5th and 97.5th percentiles of 1,000 row-level bootstrap draws from their observed binary confusion counts [29]. These are descriptive within-panel intervals, not patient-level uncertainty. The later SPH studies report component intervals for their own contrasts.

Table 10. Metric definitions used in the manuscript package.

Metric	Definition	Use
Top-1 accuracy	Correct primary class predictions divided by evaluated records	Multiclass rhythm
Mean binary accuracy	Arithmetic mean of one-vs.-rest class accuracies	Multilabel pathology
Macro-F1	Unweighted mean of classwise F1 scores	Both tasks
AUROC	Area under the receiver operating characteristic curve	one-vs.-rest
Average precision	Recall-weighted precision at distinct score thresholds	one-vs.-rest
Brier score	Mean squared error between predicted probabilities and binary outcomes	Probability calibration
ECE	Ten-bin expected calibration error	Probability calibration

2.7.1. Probability calibration and decision thresholds

The binary reliability analysis retains the original binning formulation, substituting reference-positive frequency for multiclass correctness:

$$\text{ECE}_{\text{binary}} = \sum_{b:n_b>0} \frac{n_b}{n}|\bar{y}_b - \bar{p}_b|. \tag{21}$$

Here n_b is the number of observations in probability bin b, $\bar{y}_b$ is its positive-reference fraction and $\bar{p}_b$ is its mean predicted probability. The analysis uses ten fixed equal-width bins. This estimator depends on binning and the selected population; low ECE on a selected panel does not establish calibration in another population.

Probability calibration and cutoff selection address different questions. Calibration concerns agreement between predicted probabilities and observed event frequencies for a specified reference and population. A cutoff defines a binary operating point. A high AUROC does not imply either calibrated probabilities or an appropriate decision threshold [30]. The retained pathology cutoffs were 0.670 for ASMI, 0.705 for LVH, 0.635 for IMI and 0.595 for ISC_. Their internal calibration observations are reported separately in Appendix A; they are not the 962 selected evaluation observations. The exact historical calibration roster and its separation from every earlier development stage were not recovered.

For a vector of logits, a temperature-scaled categorical distribution can be written as

$$\tilde{p}_{ic} = \frac{\exp(z_{ic}/T)}{\sum_j \exp(z_{ij}/T)}, \qquad T > 0. \tag{22}$$

This equation defines a calibration family; it does not imply that a new temperature was fitted in this revision. Binary membership scores and categorical rhythm probabilities must follow their actual saved activation paths. A top-1 result cannot replace a thresholded binary call when calculating precision, sensitivity or F1. Published cutoffs remain frozen for the retrospective transfer comparisons; recalibration on a target evaluation population would constitute a separate development experiment.

The historical optimization evidence contains two seed descriptions: the public V3C promoted-configuration summary records sampler seed 197813, whereas the recovered initial-run account records seed 197802. These may describe different initialization points, but the unavailable SQLite study and incomplete execution binding prevent certification of a single replayable seed chain. We retain the promoted configuration and recorded trial ranking; exact sampler replay and intermediate checkpoint lineage remain unresolved. Configured learning-rate values likewise specify a schedule rather than a measured per-step learning-rate trace.

2.7.2. Coverage, risk and the cost of withholding

Let K_i denote the endpoints with a known reference for ECG i, $a_i \in \{0,1\}$ the joint acceptance decision, and $e_{ic} = \mathbb{1}[\hat{y}_{ic} \neq y_{ic}]$. Conditional accepted risk is

$$R_{\text{accepted}} = \frac{\sum_i a_i \sum_{c \in K_i} e_{ic}}{\sum_i a_i |K_i|}. \tag{23}$$

The no-abstention risk uses the same known opportunities without the factors a_i. A policy may lower conditional risk by removing difficult observations while withholding correct decisions and true positives. To retain the original known denominator, we also report

$$L_\lambda = \frac{\sum_i a_i \sum_{c \in K_i} e_{ic} + \lambda \sum_i (1 - a_i)|K_i|}{\sum_i |K_i|}. \tag{24}$$

The coefficient λ is an explicitly chosen penalty for an unresolved known opportunity, not an empirically established clinical utility. It cannot represent the unequal consequences of all missed diagnoses, false alarms and delays. ECG coverage and endpoint-opportunity coverage are both reported because withholding one ECG may suppress several known

calls. A patient with a withheld positive is not an emitted false negative, but that unresolved positive must remain visible in the complete accounting.

The P8/P4 joint rule operates on score distance to released thresholds. It is distinct from entropy, Monte Carlo dropout, a model of out-of-distribution probability or a clinician decision. Its later P6 fallback can return only the six P6 membership endpoints. Partial output coverage counts an ECG with any returned result, whereas complete profile coverage requires all originally requested P8/P4 outputs. The two coverage measures must not be combined. No clinician review or referral outcome was measured.

Selected-development Wilson and multinomial-bootstrap intervals describe uncertainty at the observed ECG or panel-row level. They do not account for model selection, unknown historical clustering or target-population shift. Later analyses use patient or subject/exact-signal components where available. The five-seed morphology study controls three primary profile contrasts with simultaneous 98.333% component intervals and a prespecified two-percentage-point relevance margin. Per-class seed ranges describe training variability and are not confidence intervals. The earlier single-fit paired study and the later five-seed study use different designs and must not be pooled as independent replications.

2.8. Patient-held-out evaluation within CODE after supervised adaptation

CODE-15% version 1.0.0 is a source external to the original TITAN experiments, with 345,779 ECGs from 233,770 patients in the provider release [31]. The completed analysis is a patient-disjoint evaluation within CODE after supervised domain adaptation. Because the candidate used CODE training labels, its reserved CODE evaluation does not constitute independent institutional confirmation of the adapted model. It remains separate from the historical P6/P8 selected development profiles and from the unfinished independent study.

Patient hashes determined 70/10/10/10% buckets before caps and quality control (QC), with at most two ECGs per patient. Table 11 reports the admitted partitions, whose final sizes need not preserve the initial bucket proportions. The sealed audit reports zero patient-key overlap across these four partitions. Validation, calibration and test excluded patients meeting the earlier PR 6 selection predicate. The 992 training patients flagged by that predicate are not a verified intersection with the final historical PR 6 roster. Exact observed float32 waveform hashes were screened within CODE. Across all four partitions, QC excluded 41 all-zero, 745 ambiguous-duration, 52 flat-lead, 10 cross-split exact-waveform and 127 within-split duplicate observations. These pooled counts describe exclusions across partitions, rather than test-only exclusions. They do not establish absence of cross-institution duplicates or authenticate all historical TITAN training patients.

Table 11. Admitted CODE partitions after patient caps and QC. Durations describe the delivered/used windows.

Partition	ECGs	Patients	7.335 s	10 s
Train	9,869	7,808	5,924	3,945
Validation	3,088	2,426	1,881	1,207
Calibration	3,080	2,435	1,876	1,204
Reserved test	4,921	3,891	3,061	1,860

The source HDF5 stores 4,096 samples per lead at 400 Hz in the order I, II, III, aVR, aVL, aVF and V1–V6. The provider describes zero padding for variable-length ECGs [31]; the present admitted-window audit specifically found symmetric 581/581 all-lead zero borders, leaving 2,934 observed samples (7.335 s), and 0/0 or 48/48 border patterns,

for which the center 4,000 samples were used. Other patterns were excluded. Observed nonzero extent is not a measurement of device acquisition duration. The selected pipeline applies third-order Butterworth second-order-section zero-phase filtering at 0.05–45 Hz, polyphase resampling to 125 Hz and per-lead z-score normalization. Length-grouped inputs have 917 or 1,250 samples, without synthetic padding entering the candidate. Physical acquisition gain is not inferred from HDF5 amplitude. This contract differs from the reconstructed causal 0.5–45 Hz, fixed-window original path.

The adapted candidate is initialized from the V3F residual CNN–Transformer backbone and adds a five-output multilayer perceptron with LayerNorm, 128 hidden units, SiLU and dropout. Its inputs comprise the pooled embedding, seven observed-window descriptors and two duration indicators. This head is distinct from the original 512-hidden-unit specialists and their ten morphology features. Training uses only CODE train, with masked weighted binary cross-entropy, patient weighting and regularization toward the initial backbone. The last residual block, token projection, last two Transformer layers, final normalization and new head comprise 16,992,199 trainable parameters. The selected checkpoint is seed 20261112, epoch 5, job 33696. The eligible family was chosen by highest mean validation average precision. Its seed with median validation average precision was sealed before test scoring. Other source/self-supervised or loss variants were not selected.

Provider flags AF, SB, ST, RBBB and 1dAVb map to AFIB, SB, STACH, RBBB and 1AVB. Unrecognized values are masked; each final endpoint has 4,921 usable test references. These are provider reference flags without new clinical adjudication; AFIB denotes an ECG-associated statement. PVC, NSR, PAC and the P4 pathology statements are not evaluated by this five-endpoint comparison. Candidate thresholds, in the stated order, are 0.66, 0.87, 0.48, 0.44 and 0.51; they were fitted on calibration only over the 0.05–0.95 grid to maximize F1 with sensitivity floors of 0.50 for AFIB, RBBB and 1AVB. No post-hoc probability mapping was selected. Frozen P6/P8 weights and published class thresholds were evaluated on the same observed ECGs, but their feature and probability paths differ. The comparison concerns complete operating systems and does not isolate an effect of weights, descriptors, filtering or threshold adaptation alone.

All macro quantities average the same five binary endpoints equally. Average precision is non-interpolated; Brier and ECE follow the earlier definitions, with ten equal-width ECE bins. Uncertainty is summarized with 1,000 patient-cluster bootstrap draws (seed 20261112). Each draw retains all ECGs of each sampled patient, and paired contrasts use identical draws. The intervals are conditional on the selected checkpoint, thresholds and dataset. A stratum requires at least 20 positive and 20 negative patients per endpoint. Long-window SB fails this support rule, so duration-stratum macro values use different endpoint rosters and cannot be interpreted as a duration effect.

The public evidence snapshot is pinned to commit `63c74b3fde2c51f7f8cde4ad396d77a1bd16cb29`. Jobs 33768 and 33769 respectively scored the frozen systems and reduced the sealed predictions; job 33772 independently read back those same private predictions. Its receipt reports 546 point comparisons (maximum absolute discrepancy 2.22×10^{-16}) and 30 macro-F1 interval-bound comparisons (zero discrepancy). This is an implementation readback, not an independent clinical cohort. The supplied public package permits hash-chain verification and recalculation of confusion-derived metrics and arithmetic macro averages. Individual-score AUROC, average precision and bootstrap intervals are retained as authenticated aggregate results; they are not claimed as locally recomputed from private scores.

3. Results

3.1. Overall Specialist Performance

Table 12 reports the selected development evaluations. P6 made a correct top-1 assignment for 1,564 of 1,643 ECGs (95.19%) and had a thresholded mean binary F1 of 80.09%. P4 had a mean panel accuracy of 80.23% and macro-F1 of 79.35% across 962 ECG–label observations from 733 unique ECGs. The P8 development profile yielded 90.61% top-1 accuracy and 76.14% mean binary F1 across 2,193 ECGs. The profiles overlap and are not independent comparators. These selected evaluations do not estimate performance across all inputs in a clinical workflow.

Table 12. Selected development performance. Rhythm denominators count ECGs; P4 counts panel observations.

Specialist	Contract	Observations	Primary accuracy	Macro-F1	Macro-AUROC	Macro-AP
Rhythm	Primary6 multi-class (top-1)	1,643	95.19%	80.09%	96.53%	85.61%
Pathology	Primary4 one-vs.-rest panels	962	80.23%	79.35%	87.46%	87.85%

Figure 7 summarizes top-1 classification for the 1,460 ECGs with exactly one accepted class. The overall top-1 denominator also includes 183 ECGs with more than one accepted class. Figure 8 and Table 13 report thresholded binary precision, recall and F1 on all 1,643 ECGs rather than only the single-reference subset. The 1AVB F1 of 62.58% and PVC F1 of 65.64% indicate substantial residual errors; neither supports reliable clinical performance.

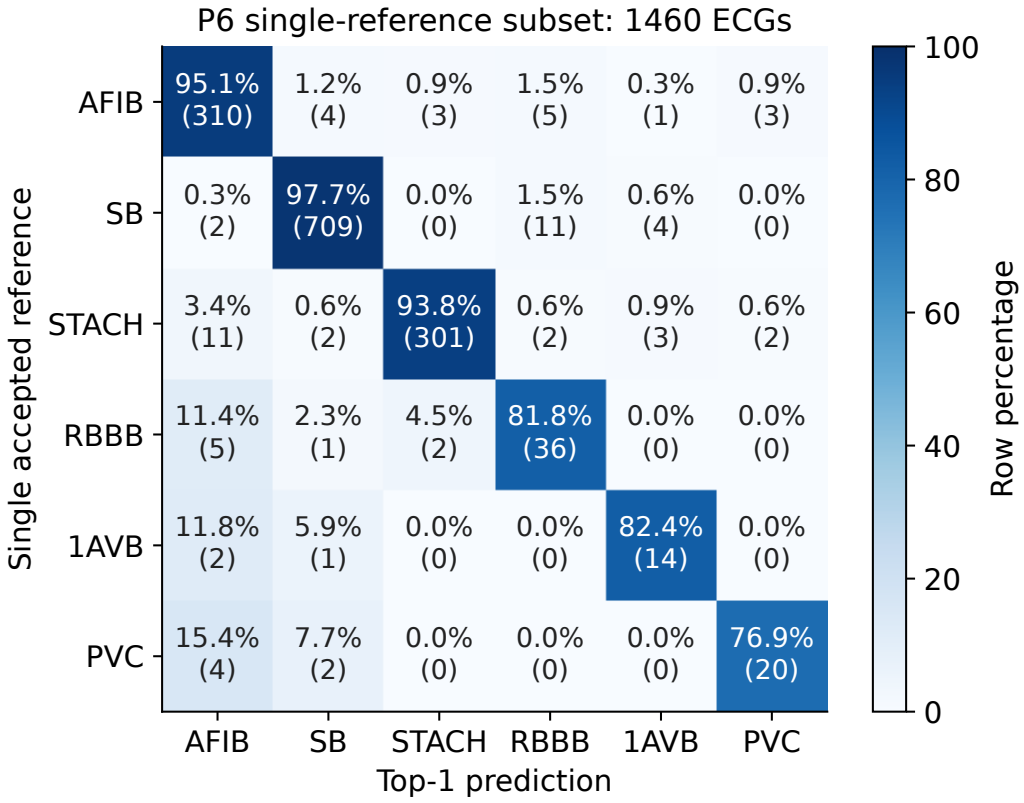


Figure 7. Primary6 top-1 confusion on the 1,460 single-reference ECGs. Cells are row percentages.

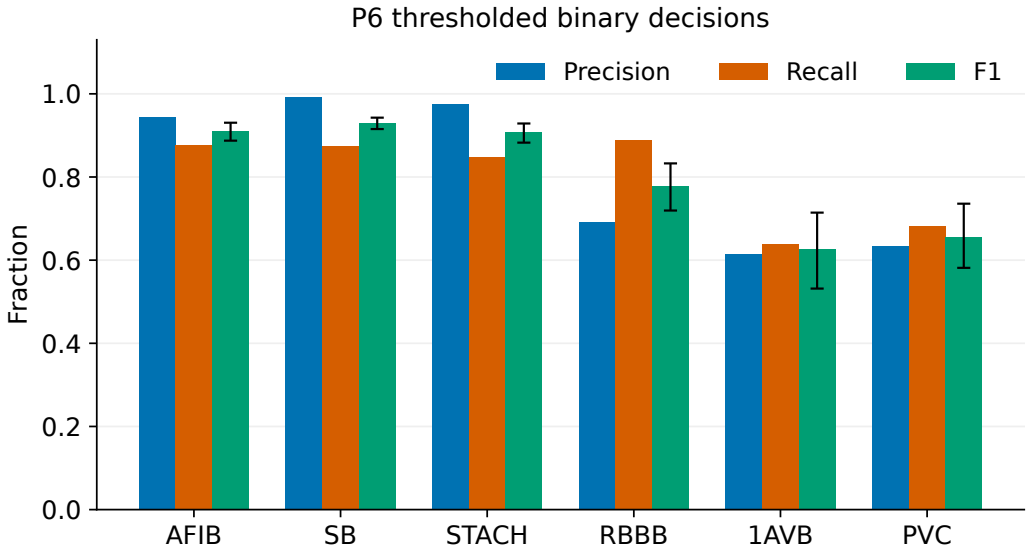


Figure 8. Primary6 classwise precision, recall and F1 on 1,643 ECGs. F1 bars include 95% row-bootstrap intervals; the plotted quantities are classwise, not macro averages.

Table 13. P6 statement-level decisions at the released thresholds. Positives are reference statements.

Class	Positive	Total	Precision	Recall	Specificity	F1	Cutoff
AFIB	390	1643	94.48%	87.69%	98.40%	90.96%	0.820
SB	792	1643	99.14%	87.50%	99.29%	92.96%	0.835
STACH	362	1643	97.46%	84.81%	99.38%	90.69%	0.775
RBBB	116	1643	69.13%	88.79%	96.99%	77.74%	0.515
1AVB	80	1643	61.45%	63.75%	97.95%	62.58%	0.570
PVC	94	1643	63.37%	68.09%	97.61%	65.64%	0.430

Table 14. P4 statement-level decisions at the released thresholds. Positives are reference statements.

Class	Positive	Total	Precision	Recall	Specificity	F1	Cutoff
ASMI	147	294	92.98%	72.11%	94.56%	81.23%	0.670
LVH	112	224	69.31%	62.50%	72.32%	65.73%	0.705
IMI	103	206	76.36%	81.55%	74.76%	78.87%	0.635
ISC_	119	238	97.17%	86.55%	97.48%	91.56%	0.595

3.2. Pathology Primary4 Results

Table 14 and Figures 9 and 10 report results for the four binary panels. ASMI, LVH, IMI and ISC_ contribute 294, 224, 206 and 238 observations, respectively. One ECG may contribute to multiple panels. ISC_ had F1 of 91.56%, compared with 65.73% for LVH; this difference does not establish performance in unselected clinical patients.

3.3. Discrimination, Calibration, and Uncertainty

Figures 11 and 12 show classwise ranking performance for P6 ECGs and P4 panel observations. The corresponding macro-AUROC values were 96.53% and 87.46%. These estimates describe different selected tasks; neither establishes calibration or clinical discrimination in an unselected population. With average precision defined as the recall-weighted sum, macro-AP was 85.61% and 87.85%. Brier scores ranged from 0.024 (SB) to 0.042 (1AVB) in P6 and 0.078 (ISC_) to 0.224 (LVH) in P4. Figure 13 compares mean predicted probability with observed positive frequency in each populated bin.

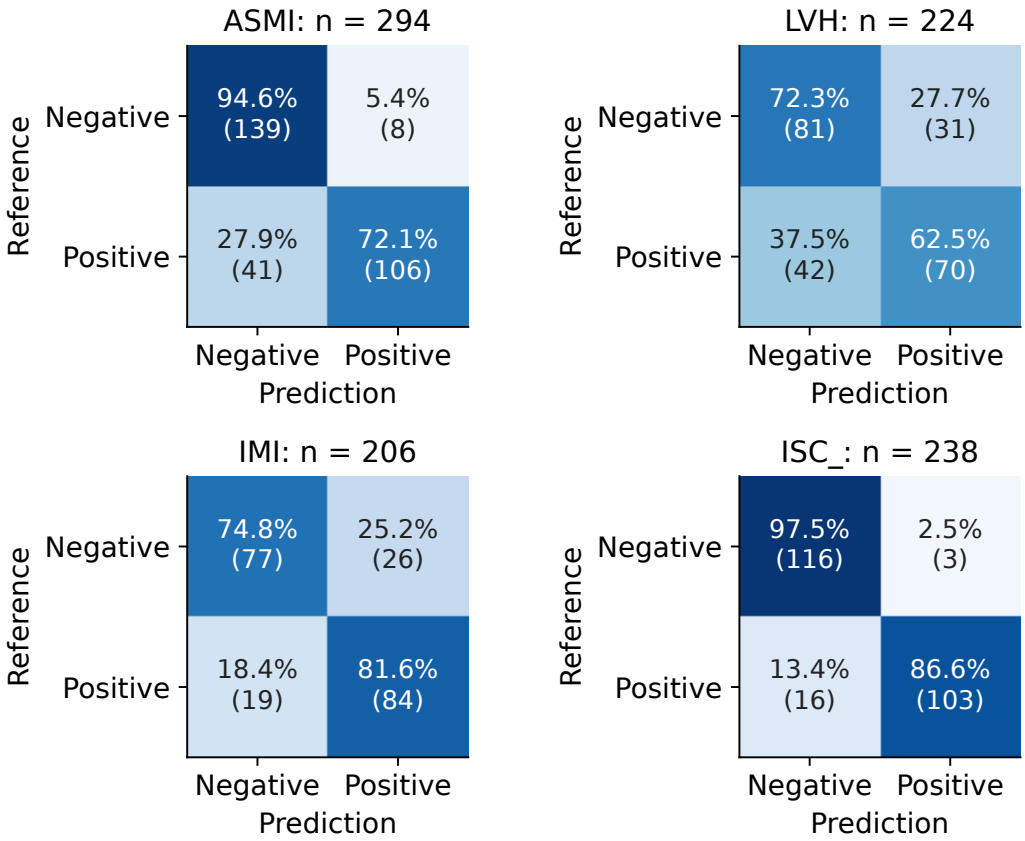


Figure 9. Primary4 confusion matrices for the four binary panels. Cells are row percentages; per-panel denominators are shown above each matrix.

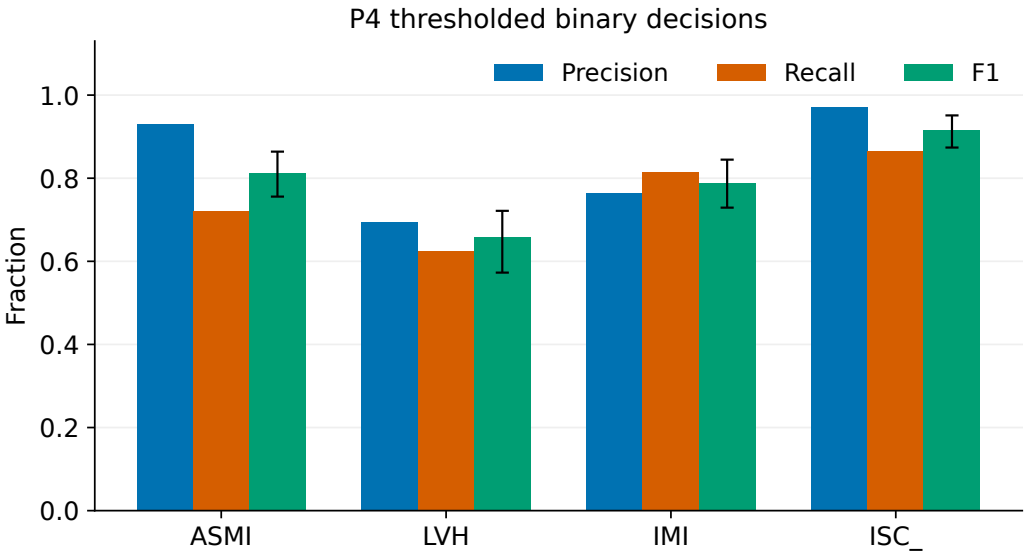


Figure 10. Primary4 classwise precision, recall and F1. F1 bars show 95% row-bootstrap intervals. These four pathology panels are distinct from the six rhythm classes.

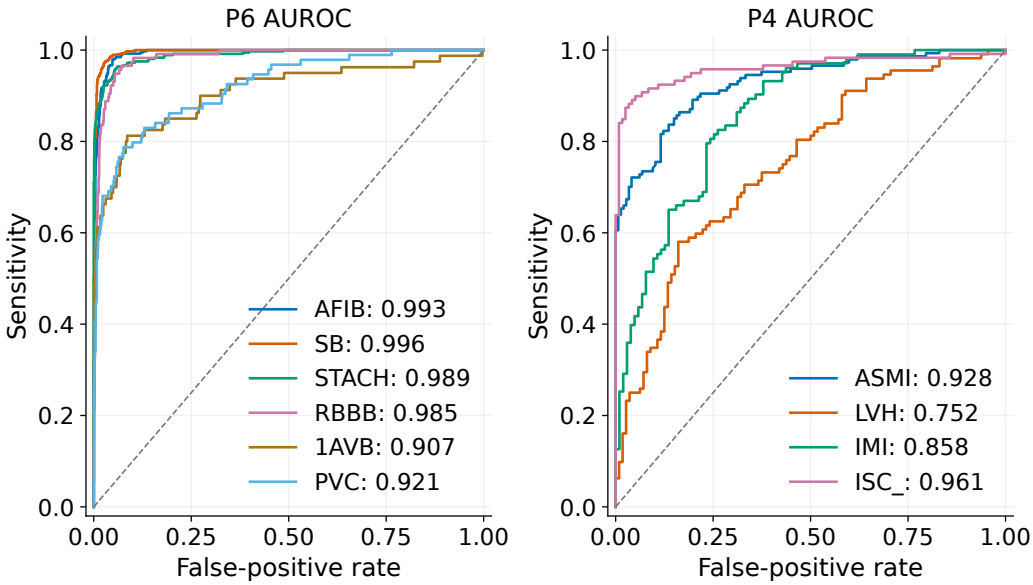


Figure 11. Receiver operating characteristic (ROC) curves. Classwise performance is mapped against the diagonal baseline for rhythm (left) and pathology (right).

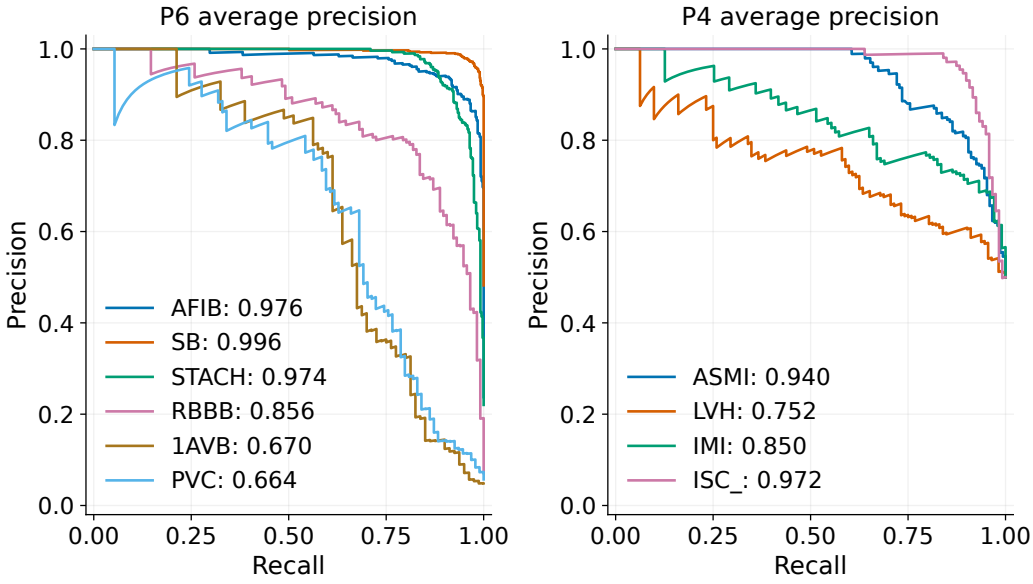


Figure 12. Precision-recall (PR) curves for rhythm (left) and pathology (right) branches. Curves document ranking quality across varying operational decision thresholds.

Threshold-empty output counts are reported separately in Section 3.7. No historical pre/post-quarantine estimate can be computed from the retained records.

3.4. Source-Stratified Robustness and Error Analysis

Figure 14 describes P6 top-1 accuracy by retained source alias. The number of ECGs varies substantially by source: Chapman–Shaoxing contributes 1,514 ECGs, while several other groups contain only five to 19 ECGs. The resulting Wilson intervals should not be interpreted as evidence of stable transport across institutions.

3.5. Denominator sensitivity in retrospective transfer

Each released profile scored all 25,770 SPH ECGs in the all-input comparison. With scores unchanged, mapped macro-F1 was 0.7083 (P6), 0.6622 (P8), and 0.0388 (P4) across

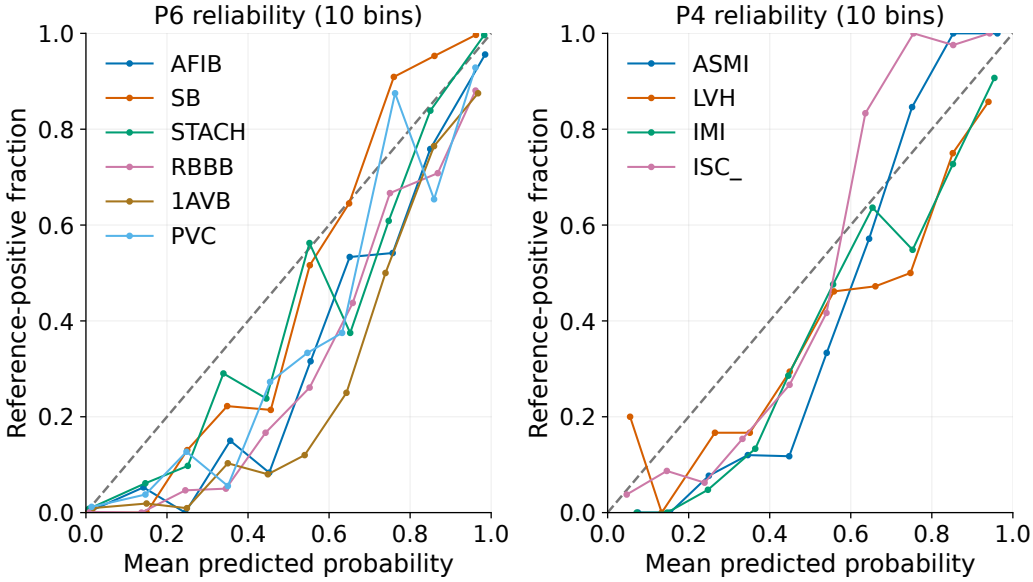


Figure 13. Reliability diagrams in ten equal-width probability bins. The vertical axis is the observed positive reference fraction; empty bins have no estimate.

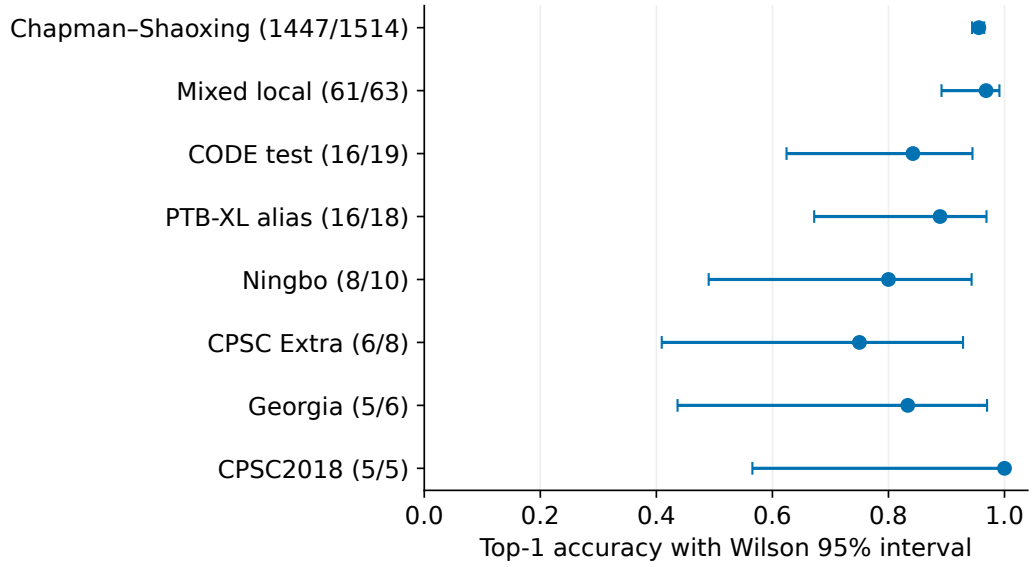


Figure 14. Source-stratified Primary6 rhythm top-1 accuracy. Variations are grouped by origin dataset family, complete with Wilson score 95% confidence intervals.

the full cohort, compared with 0.9057, 0.8553, and 0.7278 in the nested profile-compatible subsets (5,523, 5,925, and 415 ECGs, respectively). These metrics average compatible statements; they do not reproduce the original P6 top-1 task or the P4 selected panel experiment. Their interpretation depends on class prevalence and the evaluation denominator. For example, the SPH PVC reference has 1,067 positive and 24,703 absent-statement comparisons; P6 PVC F1 was 54.08% and P8 PVC F1 was 66.16%. Table 15 reports statement predictive values and patient-component F1 intervals on the original SPH denominator. These results do not supply the missing full June workflow or establish an independent confirmatory result.

3.6. Exploratory residual classes and morphology

The new rhythm residual heads yielded F1 0.647 for the P6 mapping (141 positives among 2,895 known ECGs) and 0.625 for the P8 mapping (43 among 2,797), versus 0.092

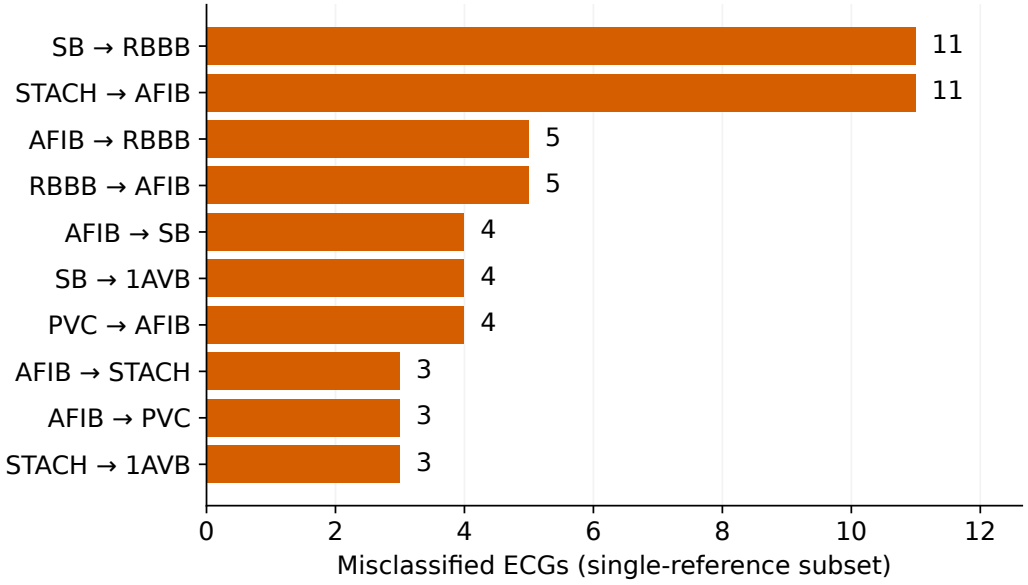


Figure 15. Frequent P6 reference-to-top-1 errors in the single-reference subset. Bars count ECGs; no bar represents a quarantine or referral.

Table 15. Retrospective SPH full-denominator mapped endpoints. Each row contains 25,770 ECG statements. F1 intervals use 2,000 patient and duplicate component bootstrap draws. Unmapped endpoints are omitted.

Endpoint	Positive	PPV	Recall	NPV	F1	F1 95% CI
P6:AFIB	675	84.86%	93.04%	99.81%	88.76%	86.92%–90.38%
P6:SB	2,711	68.41%	82.18%	97.85%	74.66%	73.46%–75.83%
P6:STACH	725	77.06%	85.24%	99.57%	80.94%	78.62%–83.08%
P6:RBBB	710	39.13%	96.62%	99.90%	55.70%	53.15%–58.12%
P6:PVC	1,067	39.69%	84.82%	99.31%	54.08%	52.02%–56.17%
P8:AFIB	675	82.29%	93.63%	99.83%	87.60%	85.65%–89.39%
P8:SB	2,711	66.04%	88.38%	98.58%	75.60%	74.35%–76.71%
P8:STACH	725	69.77%	87.86%	99.65%	77.78%	75.47%–79.95%
P8:RBBB	710	48.09%	94.23%	99.83%	63.68%	61.10%–66.04%
P8:PAC	539	16.88%	61.60%	99.13%	26.50%	24.11%–28.74%
P8:PVC	1,067	57.97%	77.04%	98.99%	66.16%	64.01%–68.35%
P4:ASMI	91	2.46%	61.54%	99.85%	4.73%	3.51%–6.09%
P4:LVH	209	1.87%	71.77%	99.67%	3.65%	3.04%–4.28%
P4:IMI	120	1.65%	89.17%	99.93%	3.25%	2.65%–3.87%

and 0.030 for the calibrated auxiliary-score comparators. A new pathology union head yielded F1 0.939 on 153 known observations out of 390 held ECGs. These are development results, not outputs of the released specialists. Paced rhythm and LQTS lack compatible SPH references, and 3AVB had insufficient support for individual fitting or calibration. A composite LAO/LAE reference does not validate either diagnosis separately.

In the five-seed fixed-backbone comparison, adding morphology changed mean macro-F1 by $+0.17$, -0.11, and $+0.31$ percentage points for P6, P8, and P4. Simultaneous intervals were $[-0.70, 1.03]$, $[-0.76, 0.60]$, and $[-0.36, 1.17]$ percentage points, respectively, within the prespecified ± 2 point margin. The experiment did not establish a practically important gain for these heads and did not test the effect of end-to-end re-training or transport to new sources.

Table 16. Different denominators in the executed LUDB route study. No row represents clinician review.

Strategy	ECGs	Known emitted	Errors	Unresolved
No abstention	200	924/924	192	0
Joint withholding	200	697/924	139	227
Joint + P6 fallback	200	847/924	147	77

3.7. Later score-based abstention and partial fallback

The original June report does not have an authenticated per-ECG quarantine ledger. Neither the original quarantine rate nor pre/post-quarantine accuracy is estimable from the recovered evidence. The archived 79/1,643 P6 and 58/2,193 P8 threshold-empty cases (64 and 22 correct top-1, respectively) remained in their corresponding top-1 denominator. Neither count is an abstention or referral rate. The separate Plan AE manifest recorded reference-based class membership and the Plan AF handoff does not supply an eligible all-input denominator or outcomes. Neither can be used to assign a historical clinical quarantine rate.

In the later SPH held development partition, the joint P8/P4 policy abstained on 2,478 of 12,869 scored ECGs (19.26%; 95% component interval 18.55–19.89%). Known binary endpoint error was 9.53% before selection and 9.00% among accepted ECGs. The controlled difference was -0.54 percentage points (simultaneous interval $[-0.63, -0.44]$). Of 347 AF-positive ECGs, 274 had a correct emitted decision, ten an emitted false negative, and 63 no decision. Across endpoints, 740 known positives and 19,674 originally correct opportunities were withheld; 337 false negatives were still emitted. At a cost of 0.25 per withheld known opportunity, full-denominator loss was 12.08%, compared with 9.53% with no abstention. A lower conditional risk is therefore not the same as a benefit on all patients.

On 200 LUDB ECGs, 48 abstentions were sent to an executed waveform-based P6 fallback. Partial output coverage reached 200/200, but full P8/P4 coverage stayed at 152/200. In 924 known opportunities, the joint route emitted 697 decisions with 139 errors; after fallback it emitted 847 with 147 errors and left 77 unresolved. On comparable fallback opportunities, seven errors were corrected and three introduced. The fallback does not return NSR, PAC or P4 results for the withheld ECGs and did not involve a clinician. Neither analysis validates the historical June policy, full-workflow safety, or clinical benefit.

The fallback's emitted false negatives totaled 103, while 67 known positives remained unresolved. With $\lambda = 0.25$, its full-denominator cost was 17.99%; its 17.36% emitted risk uses only returned known opportunities. Incremental waveform preparation plus batch-one P6 inference had a median of 0.452 s and a 95th percentile of 0.596 s on the recorded allocation, totaling 22.21 s for 48 ECGs. These times exclude model startup and the cached primary stage, and are not prospective service latency.

3.8. LUDB and PTB Diagnostic reference-bounded comparisons

The completed LUDB study scored all 200 ECGs with no technical non-scoring. Its original joint policy accepted 152 and withheld 48. However, reference availability limits which endpoints can be evaluated. Principal-rhythm metadata gave 142 NSR references, 55 other recognized rhythms and three unknowns for the P8 NSR contrast. The NSR statement does not imply that the ECG is normal in every other respect. LVH had 108 positives and 92 unknowns; the unknowns are not negative controls. PAC had seven positives and 193 unknowns. Infarct-territory and ISC_ performance was not estimable from these metadata.

Table 17. Completed retrospective PTB Diagnostic territory comparisons. All 549 ECGs were technically scored; other references remain unknown for each endpoint.

Endpoint / score	Positive	Control	Unknown	TP/FN/FP/TN	F1 (cluster 95% interval)
ASMI / released P4	77	80	392	63/14/20/60	0.788 [0.672, 0.870]
IMI / released P4	89	80	380	66/23/21/59	0.750 [0.632, 0.833]
ALMI / experimental	43	80	426	21/22/1/79	0.646 [0.523, 0.771]
ILMI / experimental	53	80	416	22/31/1/79	0.579 [0.345, 0.764]

On common endpoints with both positive and negative references, LUDB P6 AFIB/SB macro-F1 was 0.6949 on the full eligible source versus 0.8778 in the 59-ECG positive-profile subset. P8 AFIB/SB/NSR macro-F1 was 0.7075 on the full source versus 0.7407 in its 188-ECG restricted subset. These contrasts use matched endpoints and unchanged scores; they do not represent the original six- or eight-class headline metric. No P4 macro-F1 was calculated because none of its endpoints had adequate positive and explicit negative references in LUDB. The absence of a metric reflects a reference limitation rather than a zero score.

The separate PTB Diagnostic analysis retained all 549 ECGs from 290 subjects and scored them before joining reference information. Table 17 reports the subset with known references for each endpoint. Original P4 ASMI and IMI cutoffs were unchanged. ALMI and ILMI used previously fitted experimental PTB-XL heads and do not belong to the released P4 specialist. For the reported intervals, repeated ECGs were grouped by subject/exact-signal component. These source-specific retrospective analyses do not establish the independence of every historical training ancestor.

The ECGs with explicit control references came from 52 source subjects. ASMI and IMI positives came from 27 and 30 subjects, respectively; ALMI and ILMI positives came from 16 and 22. These are not four independent cohorts because healthy controls and some subject information are shared. The compatible territory comparisons are narrower in scope than a comprehensive infarction detector. LVH and ISC_ were not promoted to territory outcomes, and an unknown record was not converted to a disease-negative reference.

3.9. Earlier and later morphology experiments

The earlier fixed-representation paired-head study reported changes of $+0.72$, $+1.46$ and $+0.06$ percentage points for P6, P8 and P4, respectively. Its profile-specific 95% intervals were $[-0.40, 2.06]$, $[0.16, 2.76]$ and $[-1.63, 1.57]$ points. A later independent recalculation from the sealed aggregates reproduced the point estimates and bootstrap draws. In a separate post hoc within-group arm-swap analysis, Holm-adjusted p values were 0.5681, 0.1356 and 0.9626 across the three profiles. Although P8's original unadjusted interval excluded zero, the post hoc family-wise test did not support a robust improvement. P6 and P4 remained inconclusive in that experiment; a nonsignificant comparison does not demonstrate equivalence.

The subsequent five-seed study used equal-capacity heads and fixed backbone representations, with the prespecified relevance margin described above. Its simultaneous intervals were within that margin for all three profiles. This supports a conclusion restricted to these fitted heads rather than a universal statement that morphology is ineffective. Seed-level differences were unstable for rare endpoints: the descriptive P6/P8

1AVB F1 differences ranged from −5.38 to +4.60 points, and P4 IMI from −5.33 to +6.44. Neither range is an uncertainty interval for clinical benefit.

The reconstructed runtime was also sensitive to preprocessing. On the 16-ECG instrumented batch, removing the bandpass changed maximum P4 probabilities by 0.5157, omitting standardization by 0.8491 and dropping lead II by 0.3225. CPU versus the recorded GPU calculation changed maximum P4 probability by 0.000531, while repeated fixed-batch GPU forwards agreed exactly. These perturbation maxima describe a small technical probe rather than comparative diagnostic performance. Repeatability of one runtime therefore cannot resolve the unavailable historical loader identity.

3.10. Reserved CODE evaluation of the adapted five-endpoint system

The three systems were evaluated on the same 4,921 ECGs from 3,891 reserved CODE patients. For the candidate, this was a patient-held-out evaluation within CODE after supervised adaptation. The evaluation population, label contract and candidate differ from those in the historical selected P6/P4 analyses. Table 18 reports the five-endpoint comparison without replacing those original estimates.

Table 18. Matched five-endpoint CODE results. F1, AUROC, AP and mean binary accuracy are percentages; Brier and ECE are on [0,1].

System	F1 [95% CI]	AUROC	AP	Accuracy	Brier	ECE
Frozen P6	34.80 [32.00–37.50]	92.40	35.74	92.30	0.1048	0.1573
Frozen P8	37.04 [33.96–40.04]	93.11	39.00	91.44	0.0951	0.1427
Adapted	60.27 [56.46–63.41]	97.81	57.95	98.31	0.0149	0.0142

Confidence intervals are patient-cluster bootstrap intervals for macro-F1 only. Macro intervals for the other metrics are unavailable in the supplied aggregate snapshot.

The candidate attained a macro-F1 of 60.27% (95% CI 56.46–63.41), compared with 34.80% (32.00–37.50) for frozen P6 and 37.04% (33.96–40.04) for frozen P8. Paired candidate-minus-baseline differences were +25.47 percentage points (95% CI +22.20 to +28.83) against P6 and +23.23 points (+19.77 to +26.87) against P8. These are absolute percentage-point differences in matched five-endpoint comparisons, not relative improvements or comparisons with the original selected six-endpoint P6 result. The operating paths and threshold-fitting histories differed, so the observed differences cannot be attributed to one architectural change.

Macro-AUROC was 97.81% and macro-AP 57.95%; mean binary accuracy was 98.31%. The latter averages five negative-dominated binary comparisons; it is neither multiclass accuracy nor the proportion of completely correct ECG interpretations. On this cohort, the adapted mean Brier/ECE was 0.0149/0.0142, lower than for the frozen controls. These cohort-specific summaries do not establish calibration at a different prevalence or institution. Macro confidence intervals for AUROC, AP, accuracy, Brier and ECE were not provided and cannot be inferred from per-class intervals.

RBBB had the highest candidate F1 (82.56%), while AFIB, SB and 1AVB remained at 54.11%, 46.58% and 52.38%. The candidate missed 45 of 101 AFIB-positive and 30 of 64 SB-positive ECGs. AFIB sensitivity fell from 68.32% (P6) and 62.38% (P8) to 55.45%. For SB, false positives fell from 994 (P6) and 1,390 (P8) to 48, but true positives also fell from 46 and 56 to 34. For 1AVB, false positives increased from 27 with P8 to 73 with the candidate, and PPV fell from 49.06% to 42.97%. STACH gained PPV while detecting one fewer positive ECG than either baseline (91 versus 92). Thus, the macro-F1 gain did not translate into uniformly improved endpoint performance or sensitivity. Figure 16 reports operating-point sensitivity and PPV with their uncertainty.

Table 19. Adapted candidate performance by endpoint on reserved CODE. All metric columns are percentages.

Endpoint	Positive ECGs	F1 [95% CI]	AUROC	AP	Sensitivity / PPV
AFIB	101	54.11 [45.36–62.63]	95.99	51.25	55.45 / 52.83
SB	64	46.58 [36.88–55.17]	98.23	43.99	53.12 / 41.46
STACH	110	65.70 [58.40–71.76]	98.73	68.47	82.73 / 54.49
RBBB	133	82.56 [76.86–87.24]	98.58	81.12	87.22 / 78.38
1AVB	82	52.38 [43.01–60.18]	97.51	44.90	67.07 / 42.97

Each endpoint has 4,921 ECG references. F1 intervals use patient clusters. Sensitivity/PPV intervals and all baseline endpoint results are retained in the aggregate supplement and Figure 16.

Table 20. Adapted candidate binary confusion counts. Each row totals 4,921 ECGs.

Endpoint	Cutoff	Positive patients	TP	TN	FP	FN
AFIB	0.66	85	56	4,770	50	45
SB	0.87	58	34	4,809	48	30
STACH	0.48	103	91	4,735	76	19
RBBB	0.44	111	116	4,756	32	17
1AVB	0.51	76	55	4,766	73	27

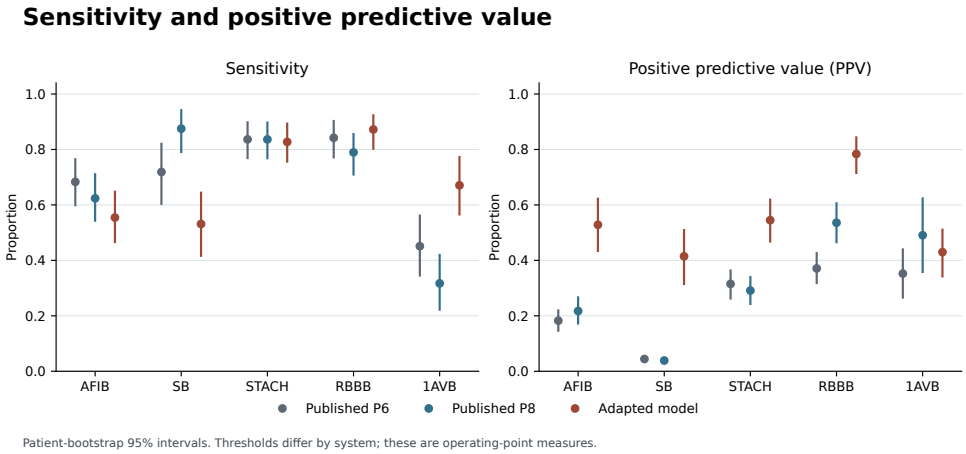


Figure 16. Sensitivity and positive predictive value for frozen P6, frozen P8 and the adapted system on matched reserved CODE patients. Points and whiskers are estimates and 95% patient-cluster bootstrap intervals. Each system uses its own frozen cutoffs; this is a comparison of complete operating systems.

Figure 17 displays the authenticated aggregate precision–recall points on the 0.00–1.00 threshold grid in steps of 0.01. Undefined precision is omitted; no interpolation or synthetic ROC coordinates are added. These sampled points do not reconstruct the exact non-interpolated AP integral. The wide separation between AUROC and AP, together with the positive support of only 64–133 ECGs per endpoint, motivates reporting both ranking summaries and integer error counts.

The prespecified research target of macro-F1 0.70 was not reached. The promotion receipt also failed the combined gain-and-safeguard rule against both frozen controls: AFIB sensitivity declined by 12.87 percentage points versus P6 and 6.93 points versus P8, and 1AVB PPV declined by 6.09 points versus P8. The candidate was not authorized to replace P6/P8. No P4 outcome or independent institutional generalization estimate is available from this evaluation.

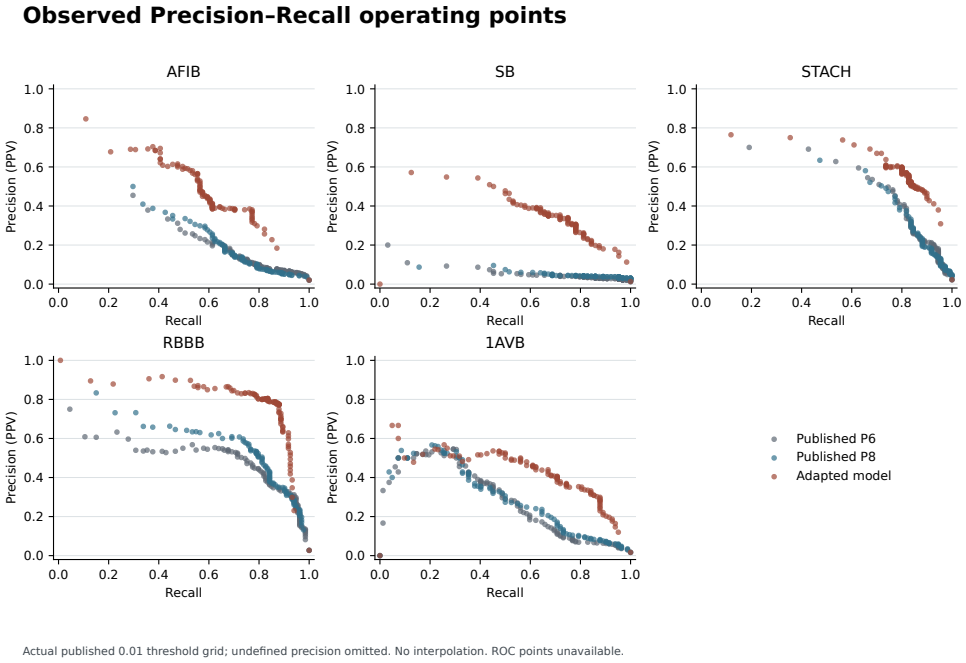


Figure 17. Observed precision–recall operating points for the same five CODE endpoints and three systems. Only the authenticated threshold-grid points are displayed. The primary receipt supplies scalar AUROC but no original ROC coordinates; this figure therefore makes no ROC reconstruction claim.

4. Discussion

The 15.10-point gap between P6 top-1 accuracy and mean binary F1 reflects different metrics and decision units, together with uneven class support. Top-1 counts a prediction as correct when the class belongs to the reference set. The unweighted binary F1 average gives each endpoint equal weight and evaluates thresholded membership. This comparison does not show that specialization eliminates bias. Interpretation remains limited by class-level errors, the concentrated source mix and the marked dependence on denominator selection in SPH.

4.1. ECG-level PVC and AF endpoints

The PVC endpoint denotes a statement associated with the ten-second ECG. It does not establish whether every ectopic beat was detected, counted or localized. An alternative design would detect PVC beats against timing-annotated references and aggregate the detected events into an ECG-level output. Such an approach would require an explicit matching tolerance, adjudicated beat labels and assessment of both beat-level and ECG-level false positives. The present study did not test that approach. AF likewise denotes an ECG-associated rhythm statement, not first-diagnosed or incident AF [17].

Tables 21 and 22 provide context from ECG studies with different tasks. Shared ECG inputs do not make their performance estimates directly comparable, because acquisition, annotation, cohort definitions and units of analysis differ.

4.2. What the retrospective transfer analyses establish

The full-source and nested SPH comparisons use the same predictions. Their large differences reflect conditioning on eligibility and the available reference ontology, with no change to the fitted weights. In particular, the P4 full-source mapped macro-F1 of 0.0388 and nested value of 0.7278 describe very different statement prevalences. A panel selected to contain comparable numbers of positive and negative references assesses whether a

Table 21. Architecture and task context across selected published ECG studies.

Study	Architecture or method	Evaluation unit and scope
Ribeiro et al.	Residual convolutional network	12-lead ECG abnormalities
Strodthoff et al.	Multiple deep learning baselines	PTB-XL multi-label benchmark
Vranken et al.	Uncertainty-aware deep network	12-lead ECG and clinical simulation
Herman et al.	Six specialized deep networks	12-lead diagnostic statements
Li et al.	Dual-view CNN–Transformer	ECG abnormalities; beat and recording features
Elsheikhy et al.	CNN with channel attention	Five beat classes; MIT-BIH and INCART
Zhou and Fang	Multiscale hierarchical CNN	Heartbeats in MIT-BIH; recordings in CCDD
TITAN	Residual CNN–Transformer	Selected ECG and binary panel endpoints

Table 22. Endpoint and evaluation units for cross-study interpretation. Scores are not a common benchmark.

Study or profile	Evaluation unit	Task
Hannun et al.	Ambulatory ECG segment	Single-lead rhythm classification
Ribeiro et al.	12-lead ECG	Six source-defined abnormalities
Strodthoff et al.	12-lead ECG	PTB-XL statement benchmark
Mastoi et al.	Heartbeat	Beat classification
TITAN P6	Selected ECG	Top-1 and binary statement F1
TITAN P4	ECG–label observation	Mean binary panel accuracy

score separates those selected groups. It does not estimate the number of false alerts generated when the positive statement is rare in an unselected source. The full-source analysis exposes this denominator sensitivity, although its interpretation remains limited by the source's statement semantics.

Source-associated reference errors can alter apparent transport independently of waveform shift. An absent diagnosis may reflect lack of assessment, different encoding or genuine absence. SPH absent-statement comparisons retain their declared mapping; unknown incidental statements in LUDB are not silently treated as negatives. PTB Diagnostic provides explicit healthy controls for named infarct territories but leaves many ECGs with an unknown reference state for each territory. Because these reference structures differ, a unified all-database macro-F1 would be misleading. Reporting known reference states and source-specific endpoints is more informative than imposing a common score on incompatible labels.

These completed analyses remain retrospective. The model has already been inspected on SPH and several public resources, and some development experiments fit new heads or policies. Source novelty alone does not restore an untouched evaluation after such inspection. The independent external study remains necessary to quantify performance under a prospectively frozen analytical contract. Its eventual results will require their own cohort flow, overlap accounting, uncertainty analysis and limitations, rather than being appended as another row in the current selected-development table.

4.3. Interpretation of specialization and uncertainty

Specialist branches organize the output vocabularies and operating points. They can also clarify which endpoints were directly supervised and which were exposed only through a global head. This organizational benefit does not demonstrate superior diagnos-

tic performance over a matched single-head baseline. The literature tables describe related methods and evaluation units, but the study did not make a controlled common-cohort comparison against all cited systems. Parameter counts describe the scale of the executed release architecture; they do not establish accuracy per parameter, energy efficiency or deployment suitability.

The two fitted backbones also limit the meaning of a shared representation. The rhythm and pathology packages implement the same backbone design, but the released V3F and V3AG files contain separately fitted parameters. They do not represent one joint forward pass with a common embedding for every reported result. A future deployment comparison should measure peak memory, startup, waveform preparation and complete latency for the actual combined route. The small LUDB fallback timing excludes model startup and the cached primary-stage runtime, so it cannot establish prospective throughput or a service-level response guarantee.

Confidence-based withholding should be evaluated with the same care as classification. The SPH policy reduced conditional known-endpoint error among accepted ECGs, while withholding correct opportunities and positive references. Its larger full-denominator loss at the selected illustrative withholding cost shows why accepted accuracy alone cannot establish benefit. Different clinical costs might favor different policies, but those costs would require an explicitly justified utility model and prospective evaluation. The current score-distance policy does not incorporate emergency urgency, patient symptoms, repeat testing or clinician availability.

The executed LUDB fallback demonstrates a narrower engineering fact: a withheld ECG can be rescored through the P6 waveform path and receive some additional membership calls. It leaves NSR, PAC and all P4 endpoints unresolved for those ECGs. Reporting complete coverage as 100% would hide that limitation. The correct descriptions are complete P8/P4 coverage of 152/200 and at least partial output on 200/200, with 77 of the original 924 known opportunities unresolved. No evidence from that execution establishes referral effectiveness or clinical safety.

4.4. Reproducibility at distinct levels

Archived predictions permit independent recalculation of selected metrics and figures, while published checkpoint lengths and hashes identify the released files. Instrumented executions establish tensor shapes and repeatability for a reconstructed configuration, and surviving run summaries describe selected optimization and training activity. None of these checks alone establishes the complete sequence of data, software, intermediate states and decisions that produced the original models.

The strict P4 historical comparison illustrates this distinction. Matching all 962 threshold decisions did not make the 836 out-of-tolerance probabilities numerically equivalent. Threshold agreement describes the retained calls for that panel, but it does not authenticate the historical probability-calibration calculation or guarantee agreement near a decision boundary in other records. Likewise, a correct SHA-256 digest identifies a byte sequence; it does not establish that a particular scheduler job read those bytes or that its training patients were disjoint from later evaluation patients.

For future releases, a more complete record would bind each training job to a versioned source tree, resolved configuration, dependency versions, patient/signal-component rosters, input checkpoint digests, optimization state, output digests and evaluation ledger. A protected record-level ledger can coexist with publicly shared aggregates and code, subject to source licenses. Such a record would help distinguish numerical replay, retraining reproducibility and independent scientific replication. The

present revision reports the available verification level and retains unresolved lineage as an explicit limitation.

Separate corrective and autonomous development records were inspected to assess whether an experimental candidate had replaced the original release. The recorded decisions retained the incumbent and did not authorize promotion. These studies used different candidate constructions or preprocessing, or repeatedly inspected selection banks. Their optimization scores are therefore not combined with the selected P6/P4 estimates or presented as independent external results. The five original released artifacts remain the model identities of the primary analysis.

4.5. Adaptation on an external source and the remaining validation gap

The completed CODE analysis provides a matched patient-held-out comparison within CODE after supervised adaptation, which the historical selected profiles do not supply. The gain is conditional on supervised adaptation, length-aware preprocessing, a new head and calibration-selected operating points. Because these components changed together, the comparison does not isolate the causal contribution of Transformer attention, morphology or one preprocessing choice. Frozen P6/P8 controls provide reference systems evaluated on the same ECGs; they do not constitute a comparison with a matched adaptation budget.

The five provider flags limit the scope of the experiment. It includes no PVC evaluation, full P6/P8 profile assessment, P4 assessment, new expert adjudication or clinical outcome measurement. The candidate showed sensitivity losses for AFIB and SB and lower 1AVB PPV relative to P8, despite the improved average. The failed promotion decision is therefore retained. Better mean Brier/ECE and high mean binary accuracy cannot replace endpoint-specific error review or a clinical utility study.

The independent implementation readback supports numerical consistency for one sealed prediction artifact; it does not add patients or institutions. Likewise, zero within-CODE patient-key overlap and exact waveform screening do not certify all historical or cross-database separation. The earlier PR 6 CODE screening involved a different cohort (24,682 ECGs, 17,300 patients), preprocessing and candidate history. Its scores must not be pooled or paired with scores from this 4,921-ECG experiment, or subtracted from them as though the observations were interchangeable. The studies document distinct stages. An independent institutional cohort with a frozen model, complete overlap accounting and compatible reference labels remains necessary to assess transport of the adapted system.

4.6. Limitations

Several methodological and computational limitations constrain the interpretation of these results.

First, the retrospective development and validation cohorts originate predominantly from public international databases in Europe and Asia. Limited and incompletely characterized representation prevents a reliable demographic fairness analysis. The small historical selected Brazilian CODE test stratum and the later within-CODE adapted-model evaluation do not establish representative Latin American or independent institutional validation, and age and ethnicity coverage were not established for all historical stages. Although the centralized data pipeline records source provenance to facilitate modular dataset expansion, any revised classification thresholds must be fitted and evaluated in a separate development cohort before any proposed deployment in a different clinical environment.

Second, dynamic lead-wise z-score normalization removes absolute voltage amplitudes to maintain gain invariance across multi-center recording equipment. This can re-

move information relevant to amplitude-dependent structural conditions, particularly left ventricular hypertrophy (LVH, F1 65.73%), for which standard electrocardiographic criteria (such as Sokolow-Lyon and Cornell indices) rely on raw millivolt measurements. The causal contribution of this information loss to LVH error was not isolated.

Third, sensitivity estimates for sparsely represented endpoints are limited by sample size and variation in native sampling. Estimates for first-degree atrioventricular block (1AVB, F1 62.58%) and premature ventricular contraction (PVC, F1 65.64%) should be interpreted alongside their confidence intervals and source-specific support. Larger curated cohorts with fine-grained electrophysiological annotations remain necessary to characterize these endpoints more precisely.

Fourth, the released rhythm and pathology profiles use separate backbone checkpoints; this separation does not establish evaluation isolation. Running the two profiles separately requires more compute than a single shared pass, but comparative deployment cost was not measured in the original evaluation.

Table 23 summarizes the planned extensions, the limitations they are intended to address and their residual technical risks.

Table 23. Planned research extensions and their relationship to current limitations.

Limitation	Mitigation Strategy	Residual Risk
Normalization removes absolute voltage; its causal contribution to LVH error was not isolated	Introduce trainable, source-specific voltage calibration layers at the specialist interface	May overfit to hardware metadata, reducing generalization to unseen devices
Regional, ancestry, and age coverage are incompletely characterized	Deploy the frozen evidence package in a diverse prospective cohort with patient-level metadata	Demographic shifts may still outpace the periodic recalibration of class thresholds
Separate fitted backbones require distinct passes; combined peak memory and latency were not measured	Distill the CNN-Transformer ensemble into a single lightweight inference engine	Compression could disproportionately degrade performance on minority conduction classes
Fixed 10 s window assumption precludes continuous Holter monitoring	Adapt temporal feature processing to support variable-length streams with artifact quarantine	Increased false-positive rates due to transient baseline wander and motion artifacts

Fifth, source reference statements do not constitute a common clinical gold standard. Mapping concordance, including the 962/962 P4 SCP-key matches, establishes consistency of the retained encoding rather than independent adjudication. Some later endpoints have only positive or unknown references, precluding estimates of specificity or predictive value. Missing statements must not be converted to clinical negatives during label harmonization.

Sixth, the complete historical Optuna database, intermediate checkpoint bytes, job-consumed partitions, exact loader and original quarantine decision ledger were not authenticated. The evidence supports archived metric recalculation and release-file identity, but not exact training replay or complete historical patient/signal separation. The reconstructed morphology and inference path is explicitly distinguished from the unavailable June runtime. No historical clinician-review effectiveness is inferred from the later score-based policy or partial fallback.

Seventh, the comparisons were retrospective, several used development populations already inspected, and selected row-level intervals do not account for all development selection or unknown patient clustering. Small source strata and rare endpoints can produce unstable estimates. Prospective clinical evaluation, demographic characterization and an

admissible independent external analysis remain necessary before drawing conclusions about safety, clinical benefit or generalization.

5. Conclusions

This study designed and evaluated a modular CNN–Transformer architecture with separate rhythm and pathology specialists. The historical selected development results support P6 top-1 accuracy of 95.19% and mean binary F1 of 80.09% on 1,643 ECGs, and P4 mean binary accuracy of 80.23% and macro-F1 of 79.35% on 962 ECG–label observations from 733 ECGs. These estimates do not measure end-to-end clinical performance. Subsequent bounded analyses demonstrate sensitivity to the evaluated population, limited benefit of the tested morphology addition, and the tradeoff between emitted errors and withheld opportunities. Released bytes and selected reported metrics can be checked independently. Complete training replay, historical patient/signal disjointness and historical clinical quarantine were not established. The completed within-CODE adaptation evaluation achieved macro-F1 60.27% on five endpoints in 4,921 reserved ECGs from 3,891 patients, with paired gains over frozen P6/P8 systems but failed promotion safeguards and persistent endpoint-specific errors. This does not validate P4 or establish independent institutional transport, which remains untested.

Supplementary Materials: Appendix A includes verified development summaries, profile-specific tables and release hashes. The companion supplement contains aggregate recalculations, Supplementary Table S1 on supervision and endpoint evidence, partition-evidence states, a morphology formula/code audit and direct model-download references.

Author Contributions: Conceptualization, I.M.G.-C., D.A.-G. and P.V.-I.; methodology, I.M.G.-C., D.A.-G., A.T.-E. and F.V.-M.; software, I.M.G.-C.; validation, I.M.G.-C., E.A.G.-L., Y.K.R.-M., A.T.-E. and F.V.-M.; formal analysis, I.M.G.-C., D.A.-G., A.T.-E. and F.V.-M.; investigation, I.M.G.-C., E.A.G.-L., Y.K.R.-M., A.T.-E. and F.V.-M.; resources, I.M.G.-C., D.A.-G. and P.V.-I.; data curation, E.A.G.-L. and Y.K.R.-M.; writing—original draft preparation, I.M.G.-C., E.A.G.-L. and Y.K.R.-M.; writing—review and editing, I.M.G.-C., E.A.G.-L., Y.K.R.-M., A.T.-E., F.V.-M., D.A.-G. and P.V.-I.; visualization, I.M.G.-C.; supervision, D.A.-G. and P.V.-I.; project administration, I.M.G.-C. and D.A.-G.

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Informed Consent Statement: Not applicable.

Data Availability Statement: Release v0.1.0 and the code and permitted derived evidence are available at https://github.com/Mateo197802/titan-v45-ecg-multihead. Table A11 identifies the five verified model files. A checksum authenticates matching bytes only; it does not demonstrate training replay, patient-level separation, or authenticity of all historical artifacts. Source ECGs remain subject to source terms and restricted records are not redistributed. The source repositories include PhysioNet [32].

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

AFIB	Atrial fibrillation
SB	Sinus bradycardia
STACH	Sinus tachycardia
RBBB	Right bundle branch block
1AVB	First-degree atrioventricular block
PVC	Premature ventricular contraction
ASMI	Anteroseptal myocardial infarction
LVH	Left ventricular hypertrophy
IMI	Inferior myocardial infarction
ISC_	Ischemic ST–T change
AUROC	Area under the receiver operating characteristic curve
AP	Average precision
ECE	Expected calibration error
MC	Monte Carlo

Appendix A. Supplementary Metrics and Evidence Tables

The appendix documents historical training, fine-tuning, optimization, source composition, transitions, calibration, record schemas, and artifact identity. The tables distinguish surviving historical summaries from selected release evaluations and later experiments. Unavailable quantities are not presented as observed values.

Table A1. Archived job 16768 summary (50 epochs). Internal development only.

Recorded field	Value
Training windows	9,021
Validation windows	924
Best composite	0.598418
Reported P8 macro-F1	0.694886
Reported P4 macro-F1	0.583806
Reported global rhythm14 macro-F1	0.326219
Reported global pathology7 macro-F1	0.532517
Clinical-axis macro-F1	0.000000

Appendix A.1. Historical optimization and run trajectory

The 50 printed epochs in job 16768 are shown in Figure A1. The trajectories describe this development run only; its logged composite is not interchangeable with the V3C stable-tail ranking or with P6/P4 release metrics. The run reported a manifest-based balanced sampler and maximum sample weight of 4. The archived report separates targeted P8/P4 from global 14/7 validation and clinical-axis scores. Clinical-axis validation macro-F1 was recorded as zero, so the existence of an axis loss does not establish useful prediction. The aggregate epoch log is supplied without record identifiers. The report's checkpoint path and the subsequent job's declared initialization support the sequence, but do not authenticate the bytes of the intermediate checkpoint.

To obtain the specific files, open the v0.1.0 release or follow the linked filenames in Table A11. Download the required model file, replace `FILE` with its local filename, and run `sha256sum FILE` on Linux, `shasum -a 256 FILE` on macOS, or `Get-FileHash -Algorithm SHA256 FILE` in PowerShell. Compare all 64 hexadecimal characters and the byte length with the table. The release also provides its checksum list. Matching bytes establishes file identity, not complete historical training replay or patient separation. V3F and V3AG are distinct waveform backbones; V3P and V3Q are P6 and P8 specialist

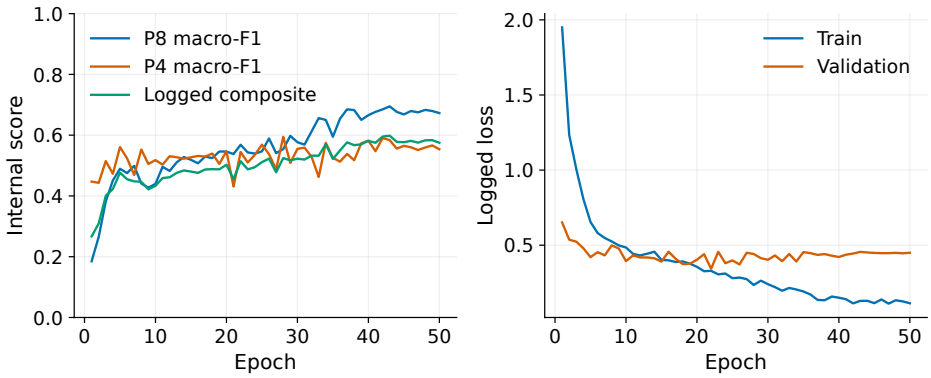


Figure A1. Internal training and validation trajectories printed by job 16768. Values retain the precision of the epoch log; the final report retains more digits. These are not performance estimates for released specialists.

Table A2. Archived V3F internal summary. Values are from the last of eight epochs.

Recorded field	Value
Rhythm validation windows	144
Pathology validation windows	776
Rhythm macro-F1	0.751823
Rhythm top-1	0.708333
P4 macro-F1	0.518821

Table A3. Archived internal P4 calibration observations and cutoffs.

Panel	Windows	Cutoff	Internal accuracy	Internal F1
ASMI	432	0.67	0.794	0.7813
LVH	374	0.705	0.7487	0.7513
IMI	492	0.635	0.7927	0.8075
ISC_	194	0.595	0.8814	0.8856

Table A4. Nine retained V3C candidates; selection by stable-tail rank, not peak.

Trial	Ranking score	Peak composite
0	0.5877250171	0.6087009381
1	0.5637407072	0.5884071929
2	0.5858734322	0.6483309166
3	0.5996263228	0.6139392727
4	0.5820539109	0.6472270611
5	0.5822570086	0.5923444733
6	0.6245117257	0.6313837131
7	0.5635969384	0.6402737567
8	0.5831962899	0.6509403856

Table A5. P6 selected ECGs by retained source family. Wilson intervals use ECGs, not patients.

Source family	ECGs	Correct	Top-1	Wilson 95% CI
chapman_shaoxing	1514	1447	95.57%	94.42% to 96.50%
codetest	19	16	84.21%	62.43% to 94.48%
data_test	63	61	96.83%	89.14% to 99.13%
training/cpsc_2018	5	5	100.00%	56.55% to 100.00%
training/cpsc_2018_extra	8	6	75.00%	40.93% to 92.85%
training/georgia	6	5	83.33%	43.65% to 96.99%
training/ningbo	10	8	80.00%	49.02% to 94.33%
training/ptb-xl	18	16	88.89%	67.20% to 96.90%

Table A6. P4 retained source alias. Pooled observation accuracy is not the mean of the four panel accuracies.

Source alias	Observations	Correct	Pooled accuracy
PTBXL_FULL_100HZ_DERIVED	962	776	80.67%

Table A7. Scope of routing evidence. Historical June entries are not estimable.

Analysis	Unit	Outcome
June historical quarantine	ECG	Not estimable
SPH later joint policy	ECG	2,478/12,869 withheld
LUDB later P6 fallback	ECG	48/48 routed to P6
LUDB unresolved after fallback	Known opportunities	77/924

Table A8. Verified decision/calibration reference and unavailable historical settings.

Setting	Supported value or location	Scope
P6 class cutoffs	Table 13	Released binary membership
P8 class cutoffs	Table A9	Includes documented empty-set filling
P4 class cutoffs	Table 14	Released binary panels
Later P8 policy margin	0.037162775	SPH calibration, profile-level confidence
Later P4 policy margin	0.014417143	SPH calibration, profile-level confidence
June class temperatures, entropy and quarantine margins	Not authenticated	Original numeric entries withdrawn

Table A9. P8 statement-level decisions at the released thresholds. Positives are reference statements.

Class	Positive	Total	Precision	Recall	Specificity	F1	Cutoff
AFIB	390	2193	92.91%	90.77%	98.50%	91.83%	0.800
SB	792	2193	94.20%	94.32%	96.72%	94.26%	0.550
STACH	362	2193	91.53%	89.50%	98.36%	90.50%	0.800
NSR	592	2193	88.30%	79.05%	96.13%	83.42%	0.525
RBBB	116	2193	74.60%	81.03%	98.46%	77.69%	0.725
PAC	103	2193	47.52%	46.60%	97.46%	47.06%	0.450
1AVB	80	2193	68.66%	57.50%	99.01%	62.59%	0.775
PVC	94	2193	65.48%	58.51%	98.62%	61.80%	0.725

Table A10. Minimum fields used to recalculate archived selected results.

Field	Use
ECG identifier	Links observations within the retained analysis
Source alias	Defines a retained source stratum, not patient independence
Reference and score	Defines binary ROC and decision comparison
Top-1 and accepted set	Defines rhythm correctness
Endpoint and binary decision	Defines pathology panel counts

Table A11. Verified release v0.1.0 artifacts. Linked filenames download the specific artifacts. Each digest covers the named file only.

File	Bytes	SHA-256
backbone-v3f-original.pt	233,565,900	`ca5d4dcb4b9828e6c4339800fe81ec27 f32d9b09a090bb57503bfb56746a8da8`
specialists-v3p-e008-primary6-diagnostic.pt	22,354,966	`9c5a9265a7155e1206c551cc4e6d645b 86363ba3dbdeb7baf7b9f173c314be3d`
specialists-v3q-primary8-best-candidate.pt	29,498,326	`b30cb2bbb45f1231c93ff1c196aa966b 57b49831cadb296131c0283e7b0ceb02`
backbone-v3ag-pathology.pt	233,566,092	`bf71e4cf8acc34031cb3611c7031e064 9822e705fd7dcea354a3ea583ce920ee`
specialists-v3ag-primary4-calibrated.pt	29,413,606	`cf5091e962fffb3254b1a71fe57236dd 82ea0139093df192511499d931391dbc`

Table A12. Later morphology study: known endpoint support and descriptive seed variability. Differences are F1 percentage points, with minus without morphology.

Profile	Endpoint	Positive	Negative	Mean difference	Seed range
P6/P8	AFIB	201	1,050	+0.59	[−2.14, 1.89]
P6/P8	SB	452	799	+0.29	[−0.02, 0.60]
P6/P8	STACH	193	1,058	+0.77	[−0.03, 1.46]
P6/P8	RBBB	50	1,201	−0.13	[−0.88, 0.63]
P6/P8	1AVB	33	1,218	−0.66	[−5.38, 4.60]
P6/P8	PVC	39	1,212	+0.12	[−1.97, 2.02]
P8	NSR	216	1,035	−0.08	[−0.94, 0.64]
P8	PAC	22	1,229	−1.77	[−5.22, 1.61]
P4	ASMI	51	135	+0.29	[0.00, 1.45]
P4	LVH	45	135	+0.06	[0.00, 0.29]
P4	IMI	57	135	+0.89	[−5.33, 6.44]
P4	ISC_	26	135	0.00	[0.00, 0.00]

weights; the remaining V3AG file is the P4 specialist. The unrecovered historical Optuna database and the generic predictions file formerly listed are not represented as verified release objects.

Appendix A.2. Five-seed morphology endpoint support and variability

Table A12 reports the known positive and negative supports for the later fixed-representation head study. P6 and P8 share six endpoint contrasts; their rows do not represent independent samples. The P4 endpoints have different known-reference masks. Counts describe eligible ECG observations rather than a verified historical training census. Mean differences average the five matched seeds; the range records the smallest and largest seed-level differences, not a confidence interval. Simultaneous profile-level intervals and the two-point relevance margin remain the primary interpretation in Section 3.6.

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